

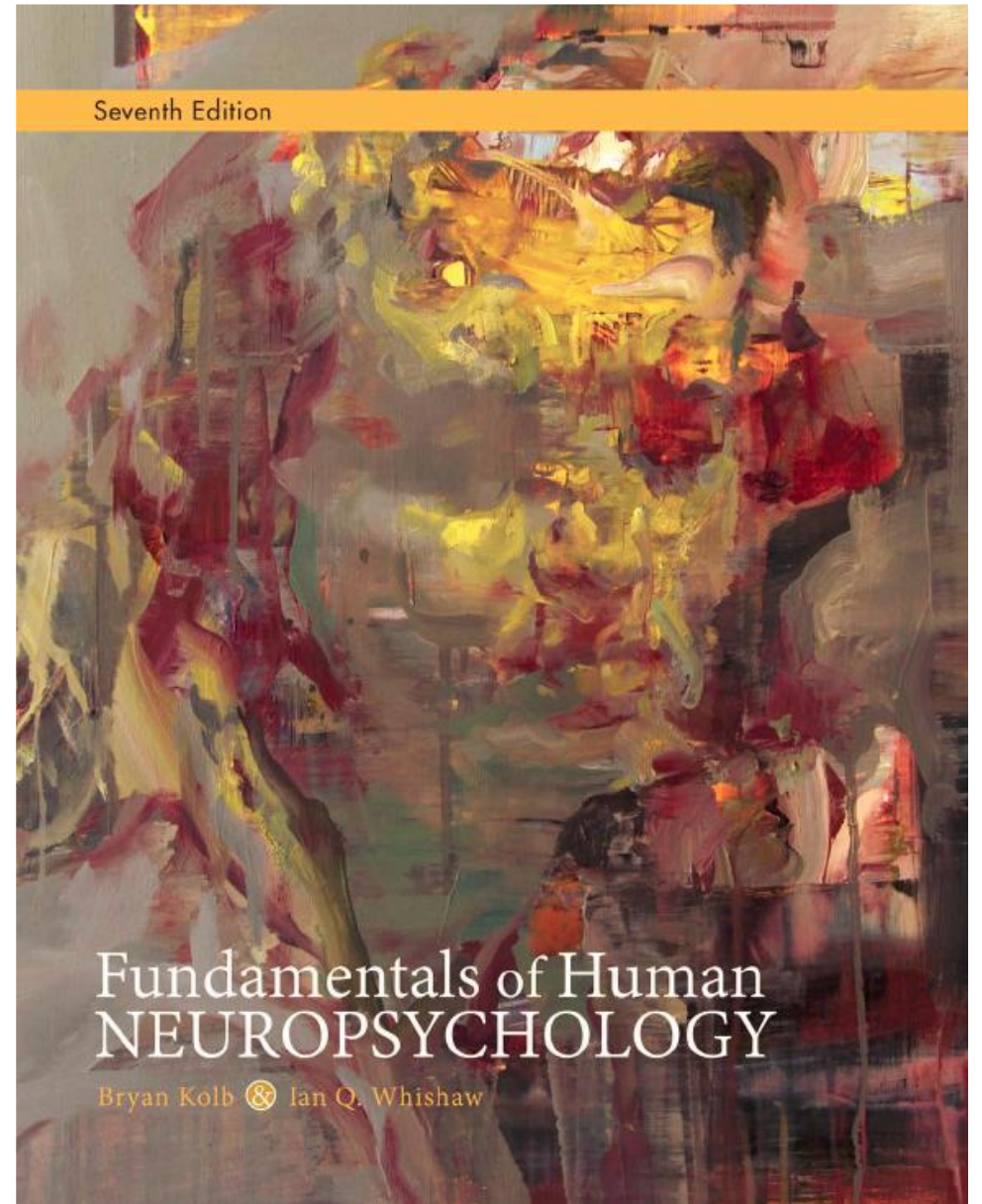
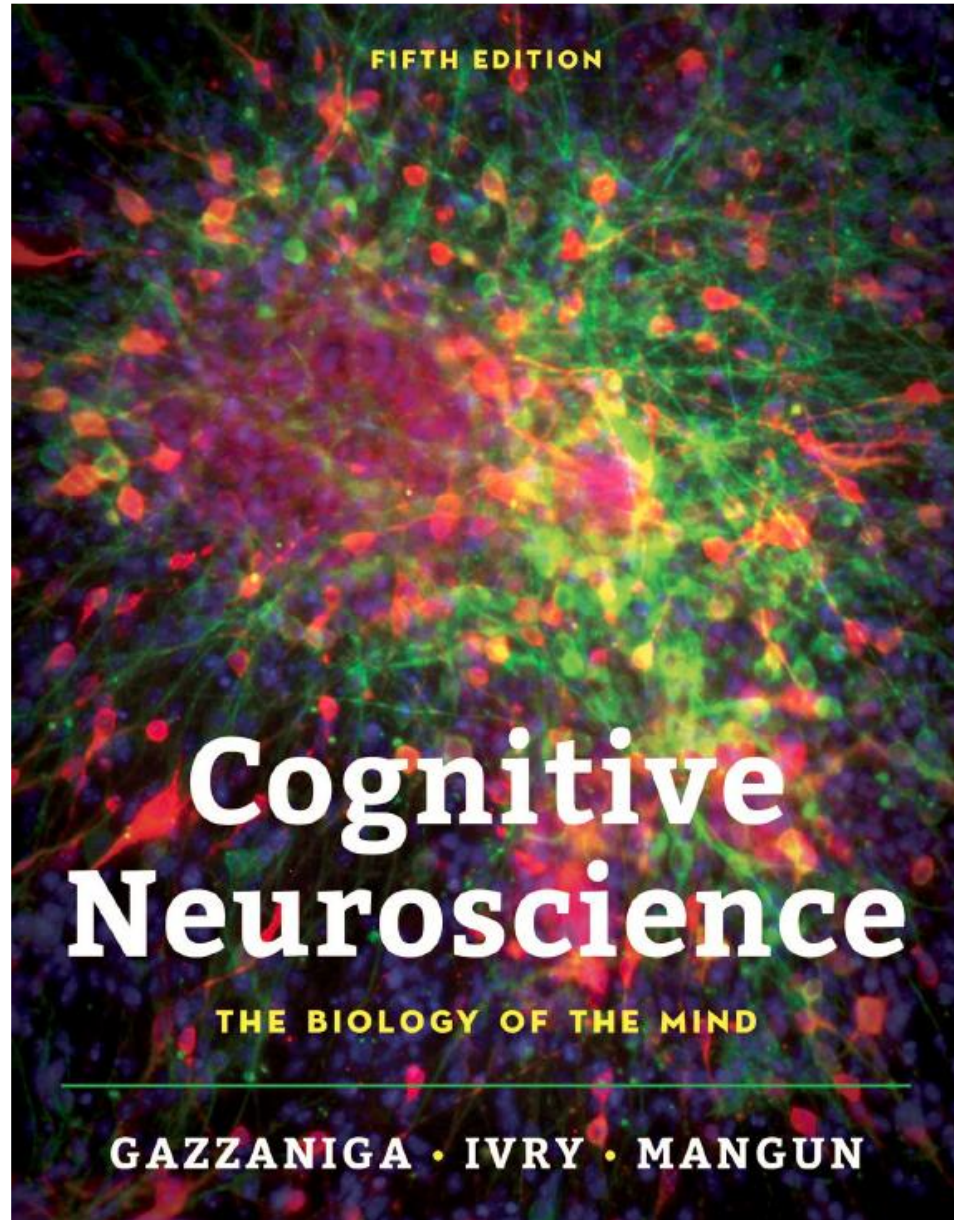
Memory and its Disorders



Axel Lindner

a.lindner@uni-tuebingen.de

Section for Neuropsychology, Hertie-Institute for Clinical Brain Research
Neurobiology of Cognition (NoCo Lab), Dept. Psychiatry and Psychotherapy
University of Tübingen





H.M.



W-H.Y.

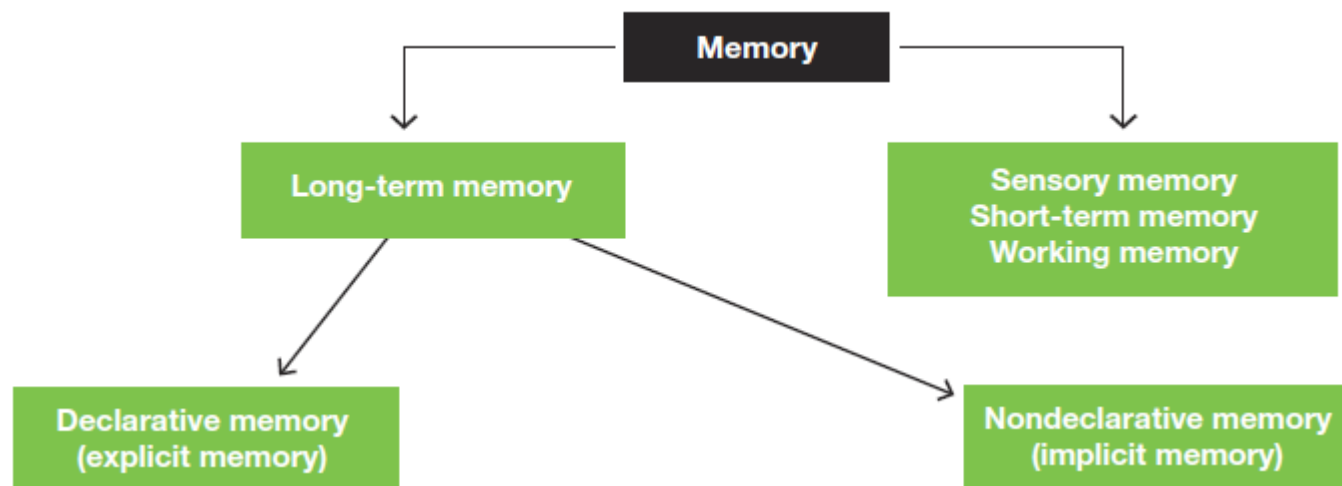


TABLE 9.1 Types of Memory

Type of Memory	Characteristic of Memory			
	Time Course	Capacity	Conscious Awareness?	Mechanism of Loss
Sensory	Milliseconds to seconds	High	No	Primarily decay
Short-term and working	Seconds to minutes	Limited (7 ± 2 items)	Yes	Interference and decay
Long-term nondeclarative	Minutes to years	High	No	Primarily interference
Long-term declarative	Minutes to years	High	Yes	Primarily interference

Why do we need memory ?

- *We need memory to maintain what we learn.*
- *It allows us to gather knowledge about the world through our experiences.*
- *This knowledge (memories) allows to optimally guide our actions in similar contexts and allows extrapolation in novel situations.*
 - > *Long-term Memory: Explicit (Declarative) vs. Implicit (Nondeclarative)*
- *Memory allows the processing of information in the absence of a stimulus.*
 - > *Working Memory*
- *Memory enables a Self (Narrative Self/Minimal Self)*



H.M.



Henry Molaison *26.02.1926 - †2.12.2008

FROM THE TIME HE WAS A CHILD, H.M. had suffered from progressively worsening epilepsy, uncontrollable by medication. By the year 1953, at age 27, H.M. was having a hundred minor seizures every day and a major seizure every few days, and he was no longer able to work.

At that time, neurologists knew that many seizures originate in the medial portions of the temporal lobe, and that their electrical impulses can spread across the brain, producing violent seizures and loss of consciousness. It was also becoming increasingly clear that surgically removing the seizure focus, the brain region where seizure activity originates, could help patients with epilepsy. William Beecher Scoville, a neurosurgeon at Hartford Hospital in Connecticut, offered H.M. an experimental surgical therapy: bilateral resection of his medial temporal lobes. Like W.J. from Chapter 4, H.M. was desperate. He agreed to the surgery. H.M.'s temporal lobes, including his amygdalae, entorhinal cortex, and hippocampi, were removed.

Although the surgery succeeded in treating H.M.'s epilepsy, his physicians, family, and friends soon realized that he was now experiencing new problems. H.M. had profound amnesia—but not the kind of amnesia that we usually see depicted in television shows or movies, in which the character loses all personal memories. H.M. knew who he was and remembered his personal history, facts he had learned in school, language, how to do things, social events, people, almost everything—up until a couple of years before his surgery. For those previous couple of years, he drew a blank. More troubling was that when a nurse left the room and returned after a short delay, he could not remember ever having seen or spoken with her before. He could follow a conversation and remember a string of numbers for a while, but he could not repeat them an hour later.



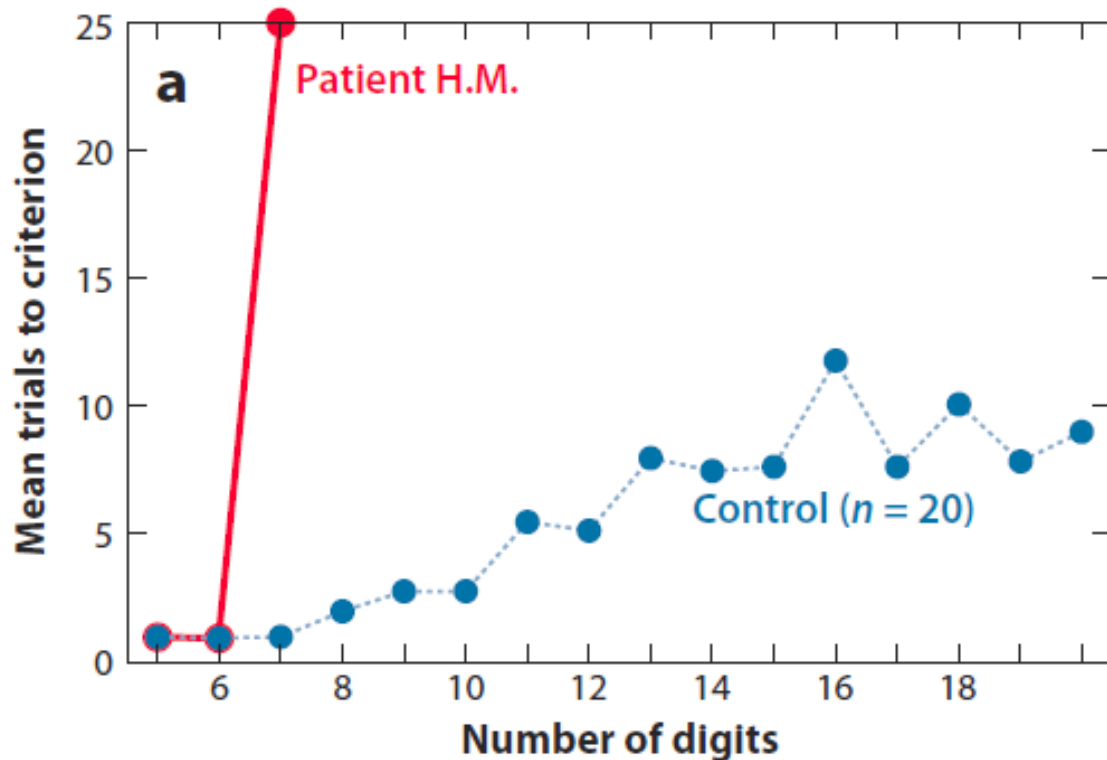
Henry Molaison *26.02.1926 - †2.12.2008

<https://www.youtube.com/watch?v=s4tF4K78BD8>

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From Squire & Zola-Morgan 2011
[Data from Dracmann & Arbit, 1966]

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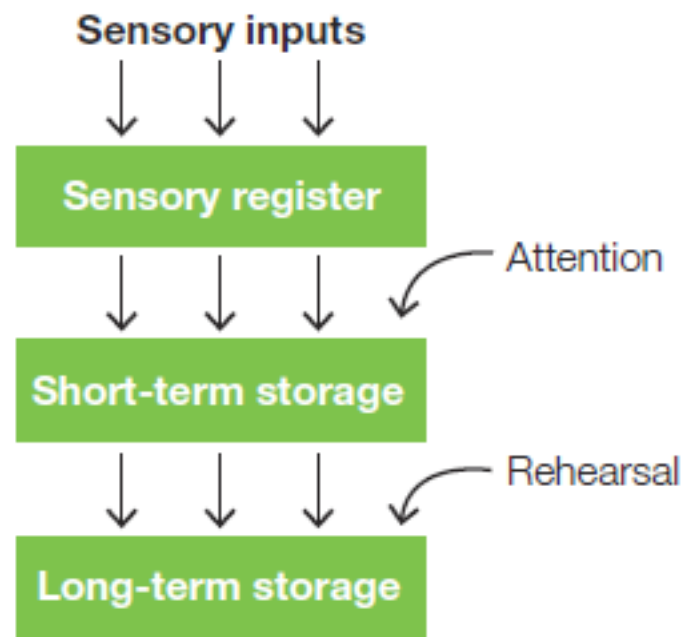


FIGURE 9.4 The Atkinson and Shiffrin modal model of memory. Sensory information enters the information-processing system and is first stored in a sensory register. Items that are selected via attentional processes are then moved into short-term storage. With rehearsal, the item can move from short-term to long-term storage.

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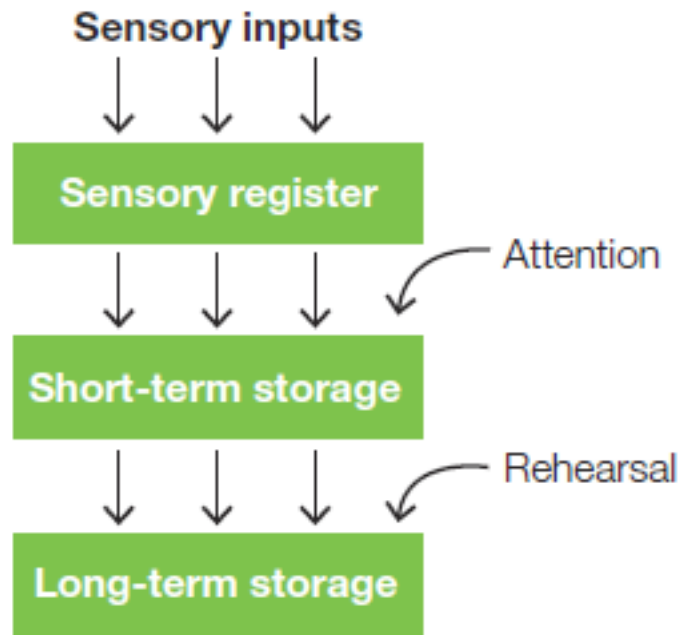
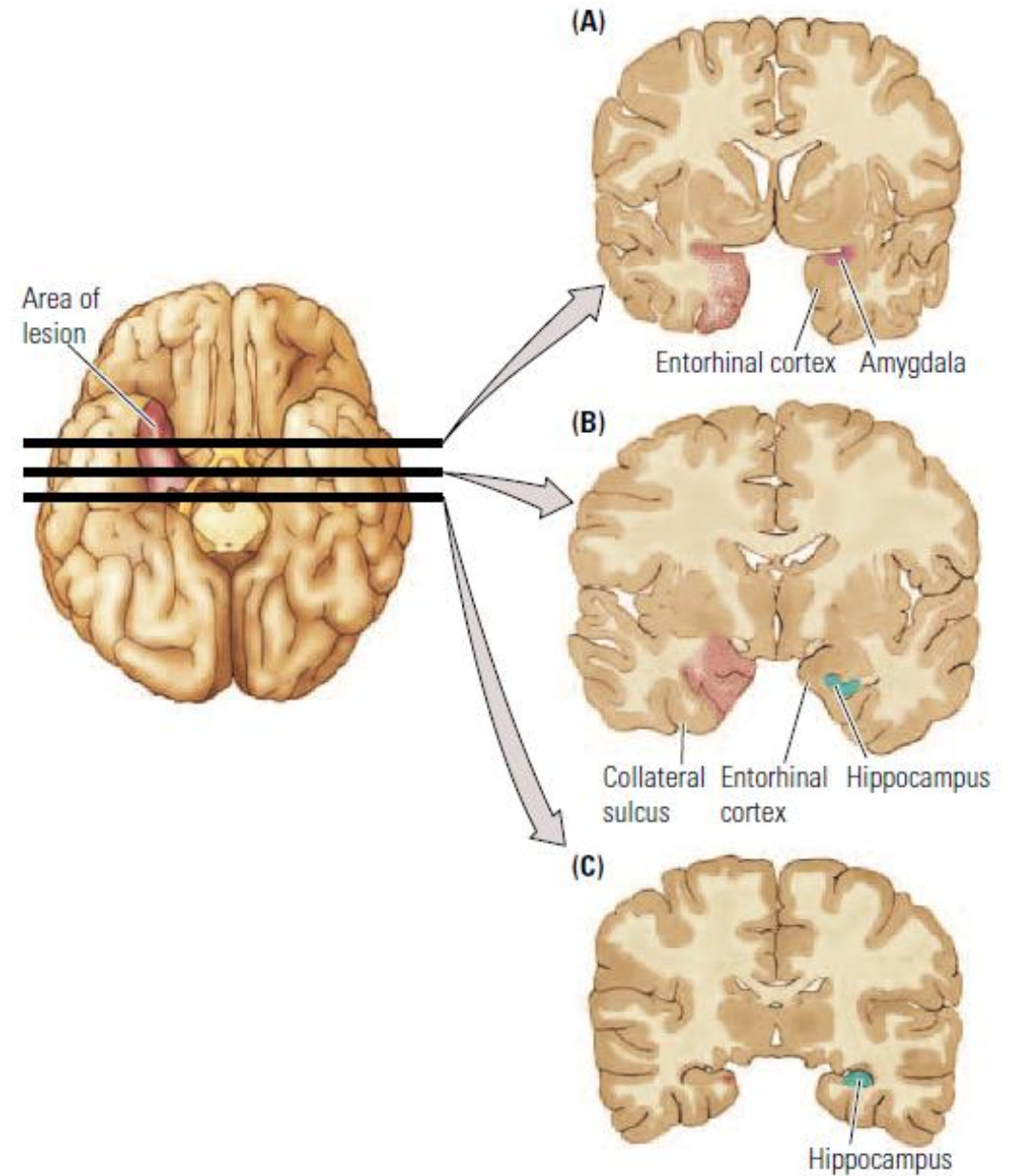


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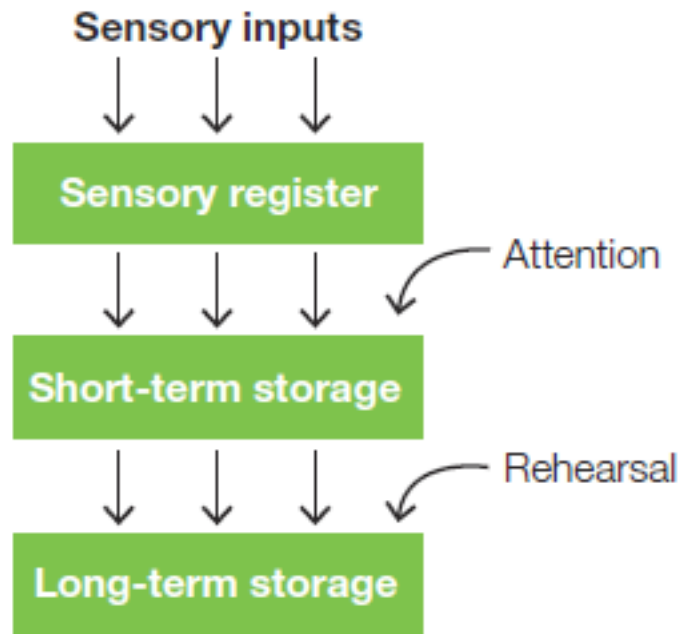
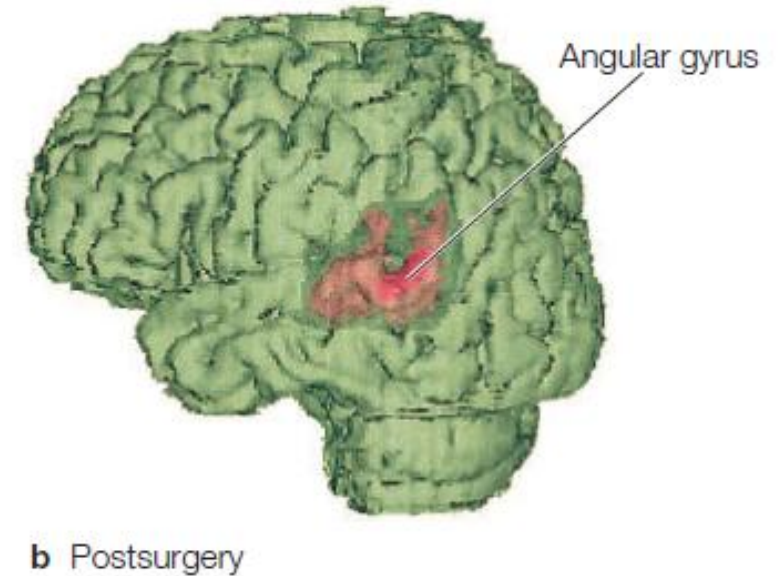


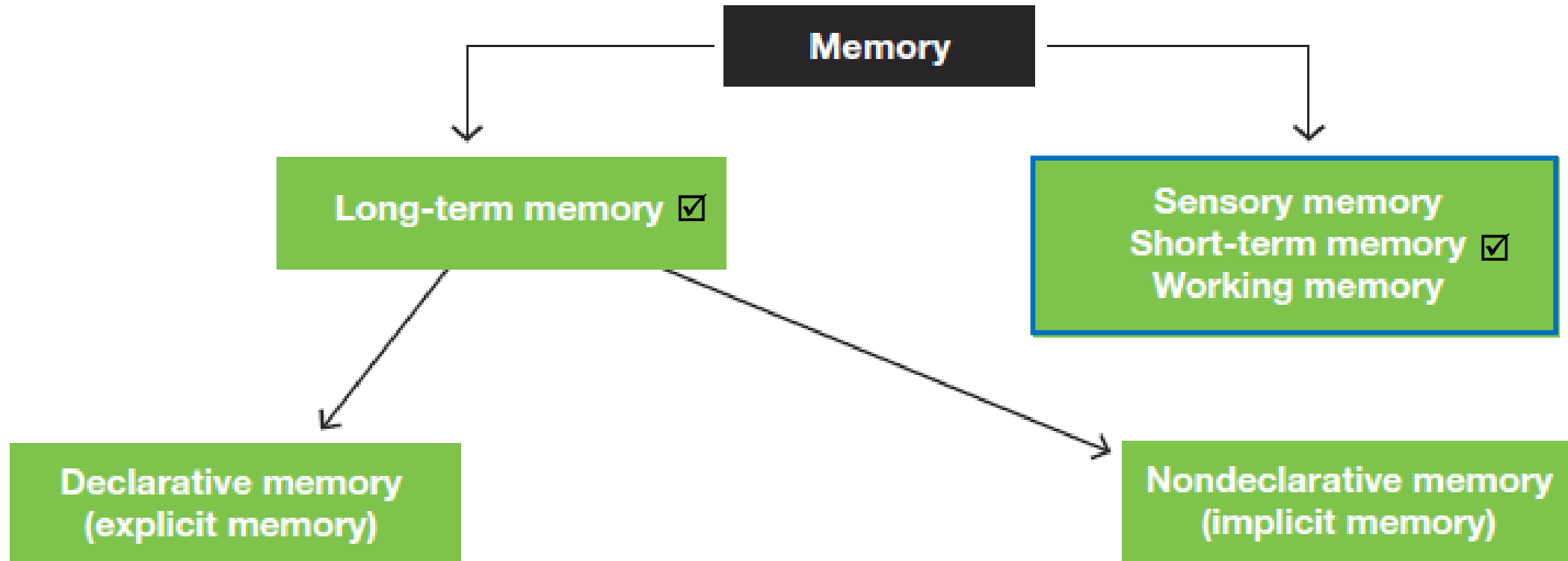
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Patient E.E. with tumor in left perisylvian cortex.

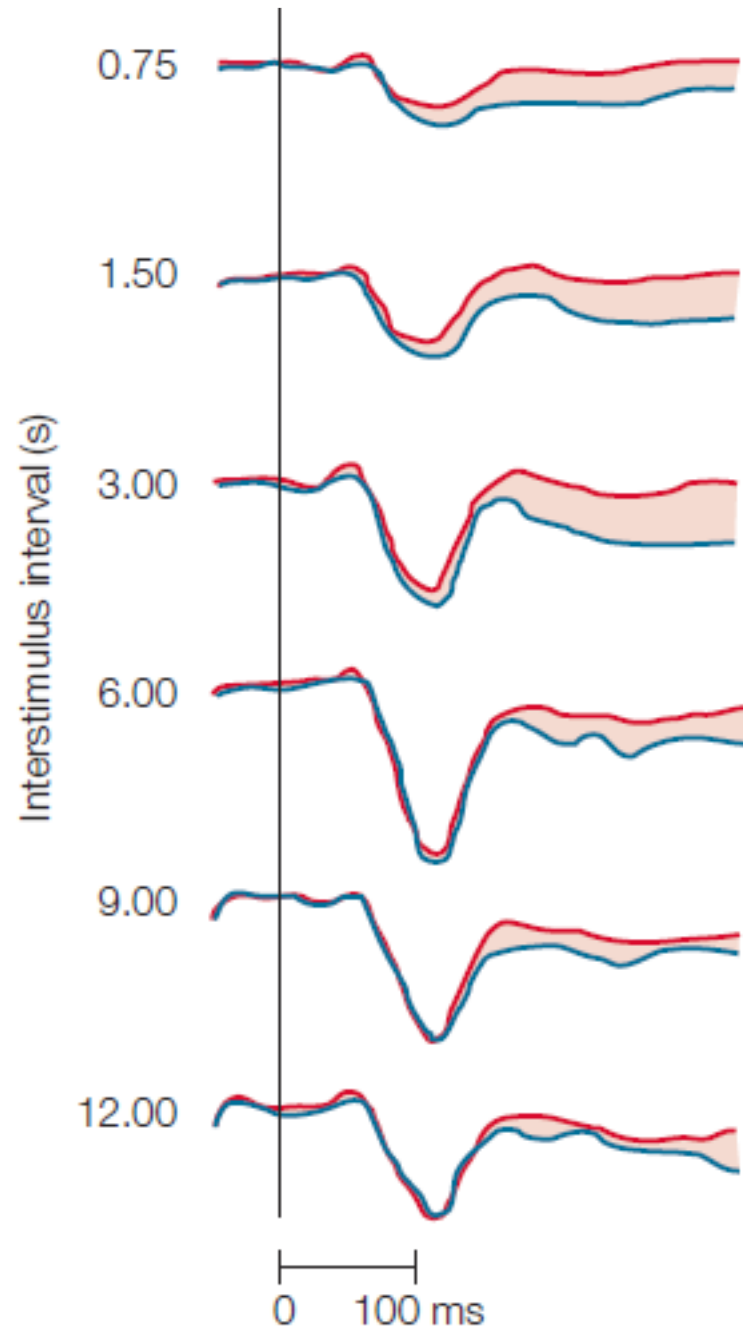
- Tumor removal caused impaired short-term memory for verbal material
- Long-term memory for verbal material was preserved

[Markowitsch et al., 1999]



Sensory Memory

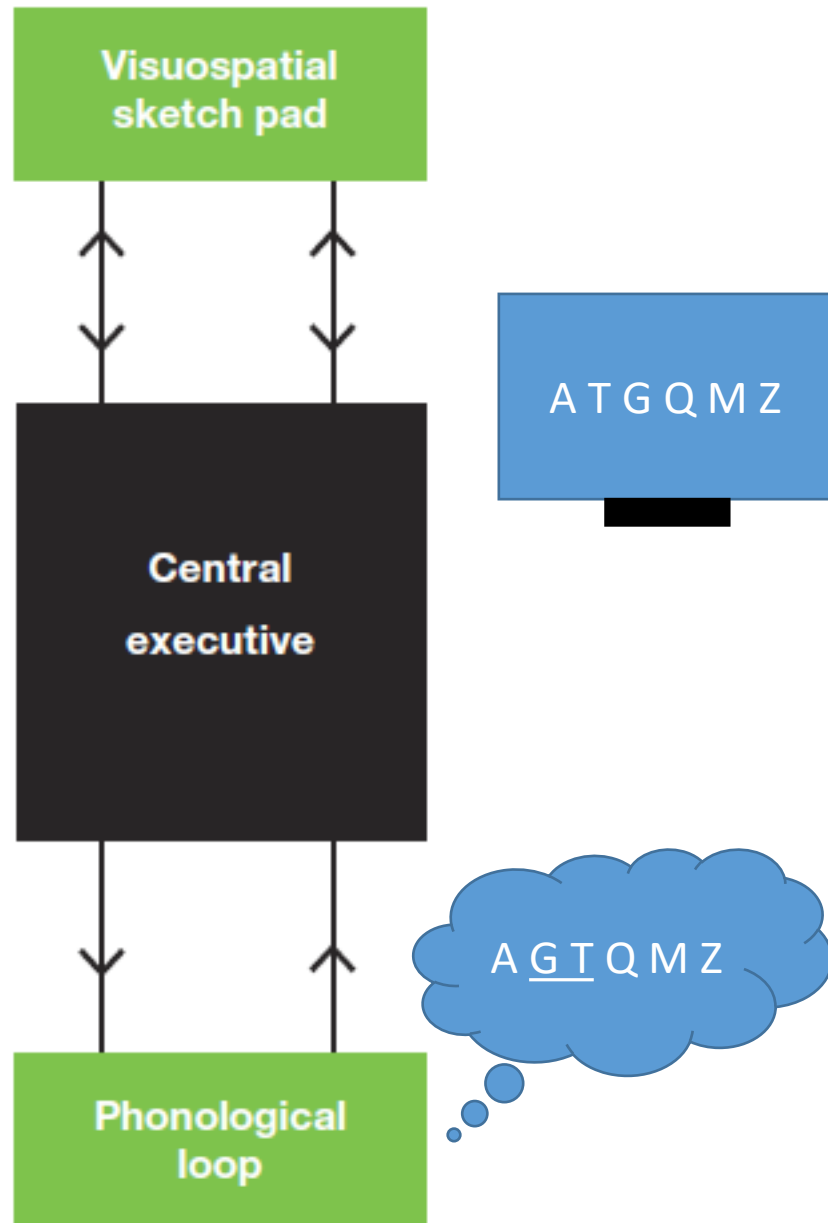
Echoic Memory / Iconic memory



The magnetic brain response known as the mismatch field (MMF) is elicited by deviant tones (blue traces) in comparison to the magnetic responses elicited by standard tones (red traces). The amplitude of the MMF (indicated by the shaded difference between the blue and red traces) declines as the time between the preceding standard tone and the deviant tone increases to 12 seconds. This result can be interpreted as evidence for an automatic process in auditory sensory (echoic) memory that has a time course on the order of approximately 10 seconds.

— Standard
— Deviant

[Ritta Hari Lab]



Working Memory

Baddeley: ,the temporary storage of information that is being processed in any of a range of cognitive tasks ‘

Goldman-Rakic: ‘the evolution of [such] a capacity to guide behavior by [mnemonic] representation of stimuli rather than by the stimuli themselves introduces the possibility that concepts and plans can govern behavior’

Cited in D’Esposito & Postle 2015, page 117

WM extends the concept of short-term memory

Verbal & Spatial Working Memory

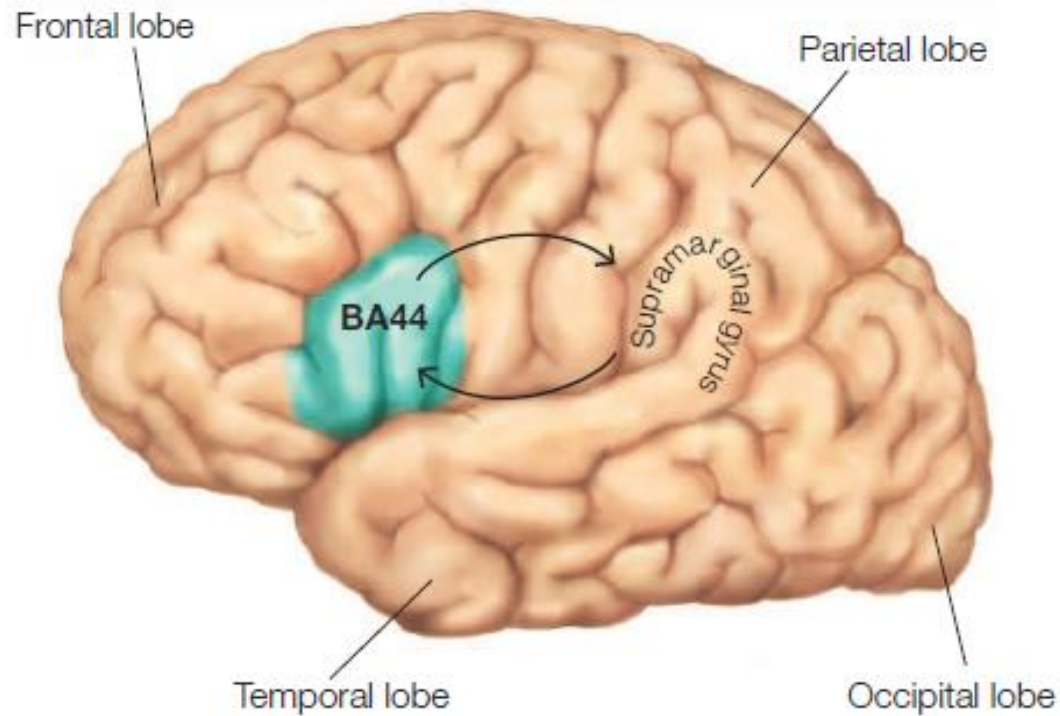
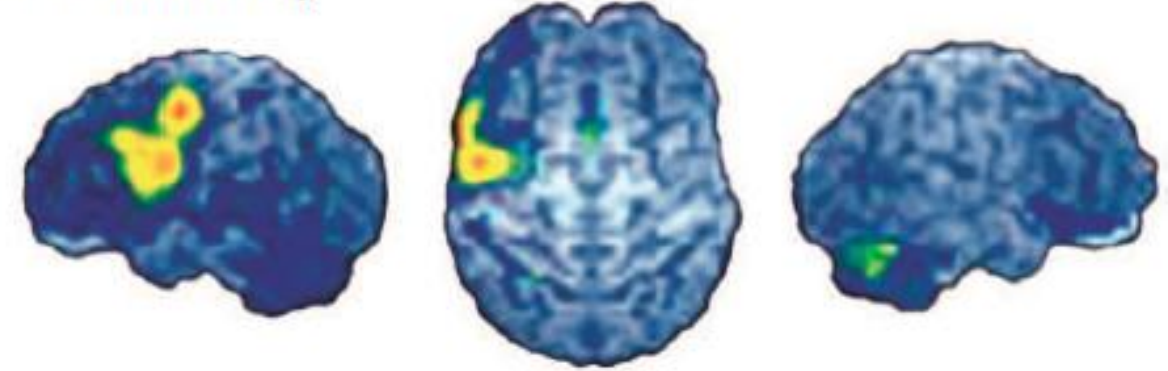


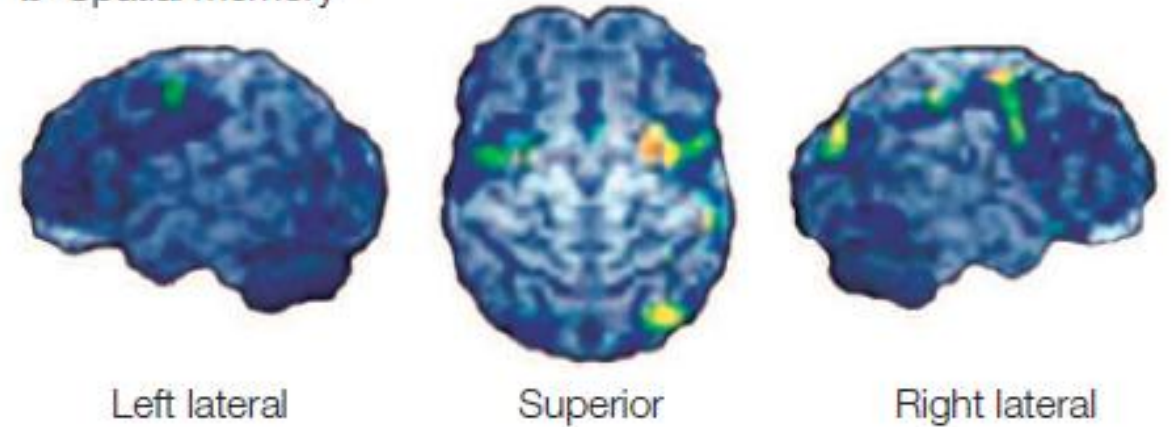
FIGURE 9.7 The phonological working memory loop. Lateral view of the left hemisphere, indicating that there is an information loop involved in phonological working memory flowing between BA44 and the supramarginal gyrus (BA40).

[Compare Patient E.E.]

a Verbal memory



b Spatial memory

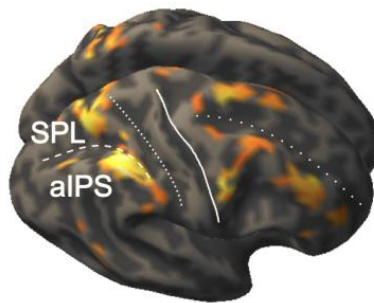
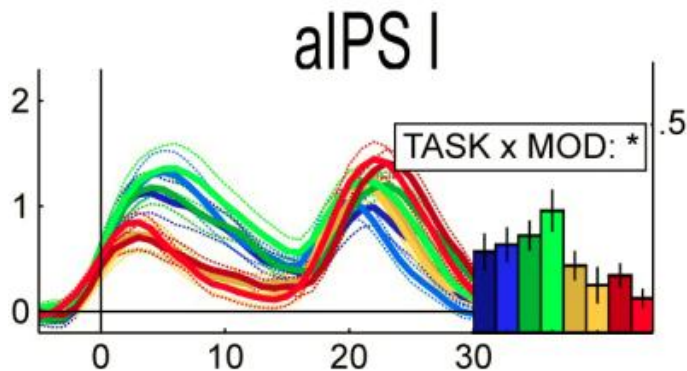
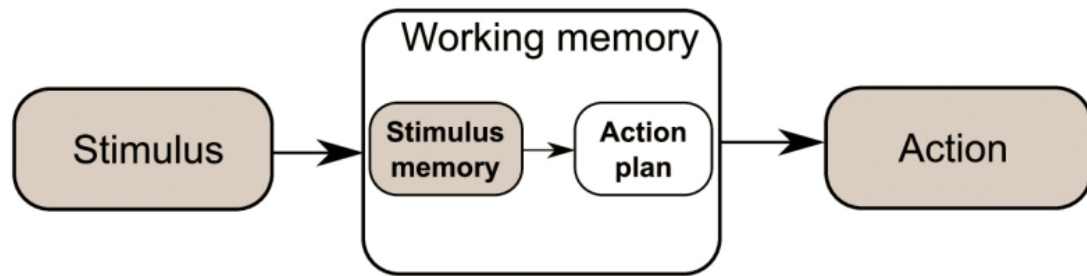


[Edward Smith Lab, Columbia; also compare Wager & Smith 2013]

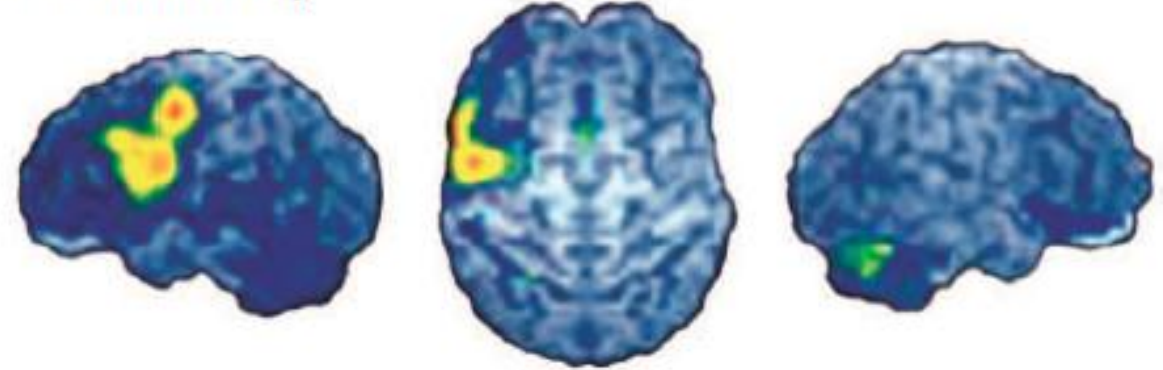
“Vocal” & “Spatiomotor” Working Memory ?

Remember how to use it: Effector-dependent modulation of spatial working memory activity in posterior parietal cortex

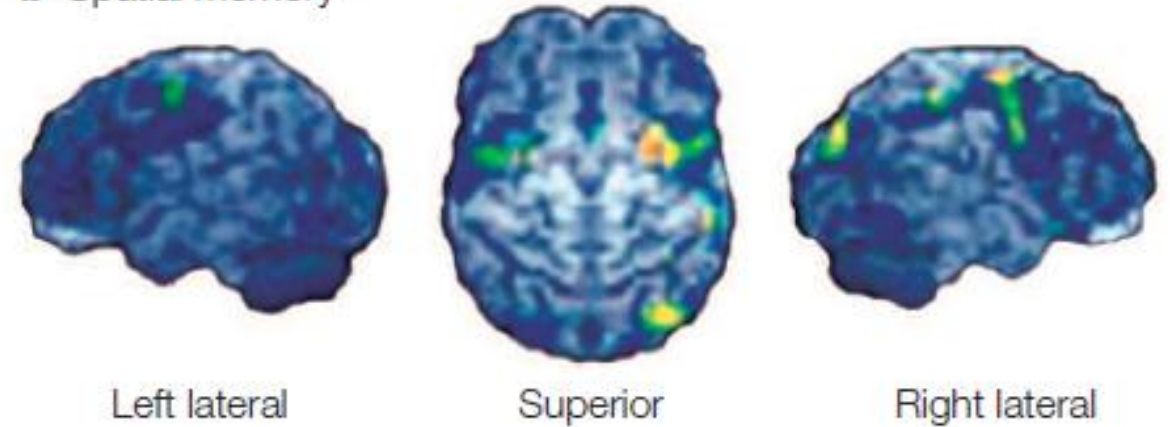
Artur Pilacinski^{1,2}*, Melanie S. Höller-Wallscheid¹, Axel Lindner^{1,3,4}* 2020



a Verbal memory

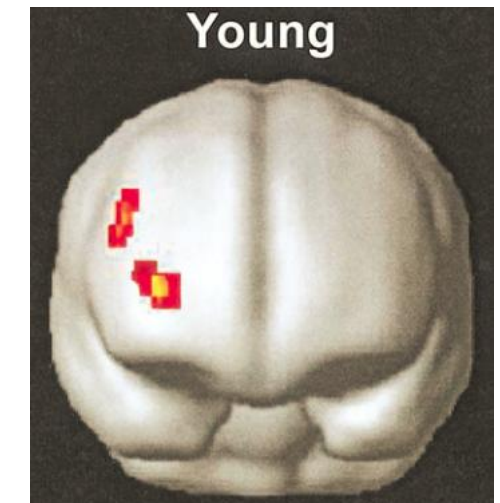
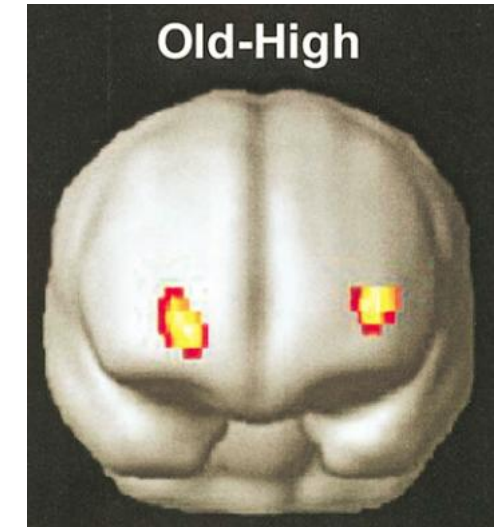
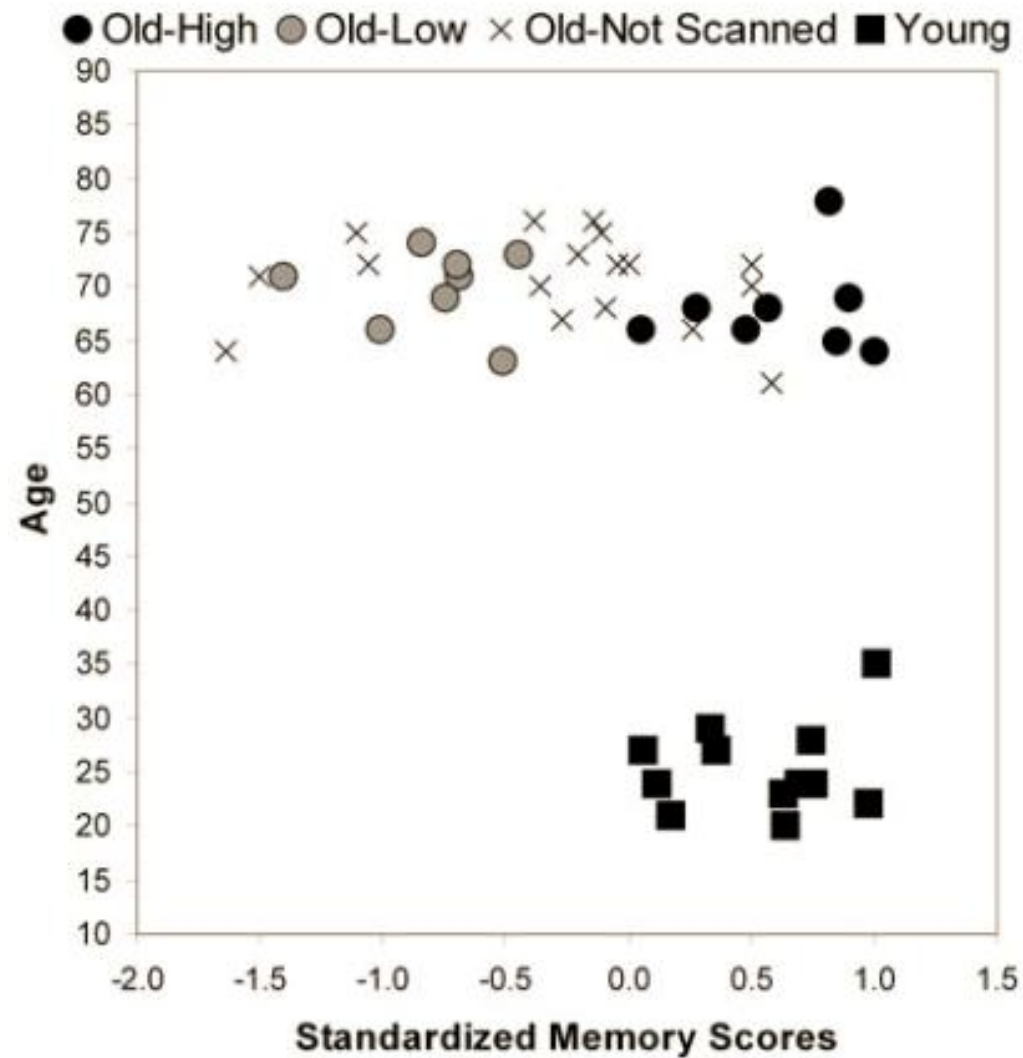
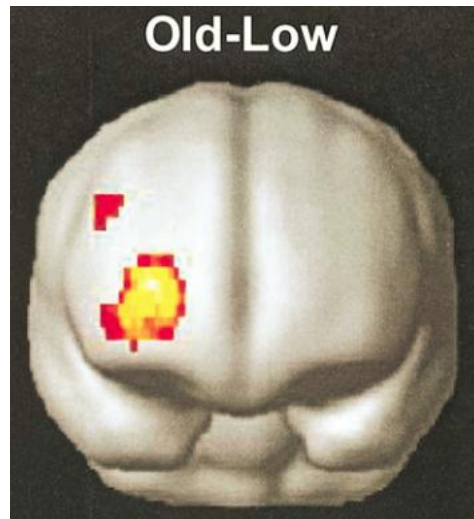


b Spatial memory



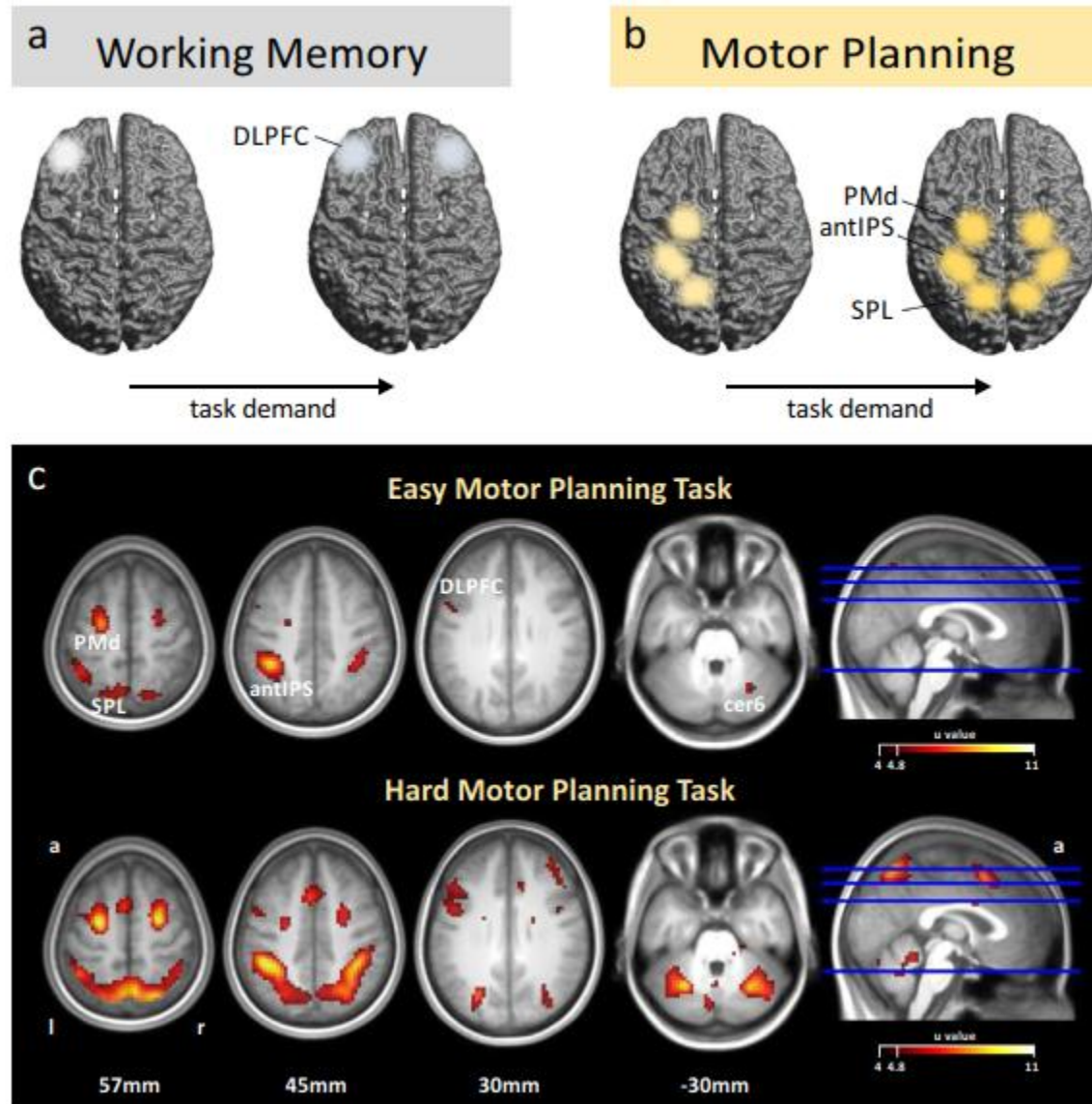
[Edward Smith Lab, Columbia; also compare Wager & Smith 2013]

HAROLD | Hemispheric Asymmetry Reduction in Old Adults



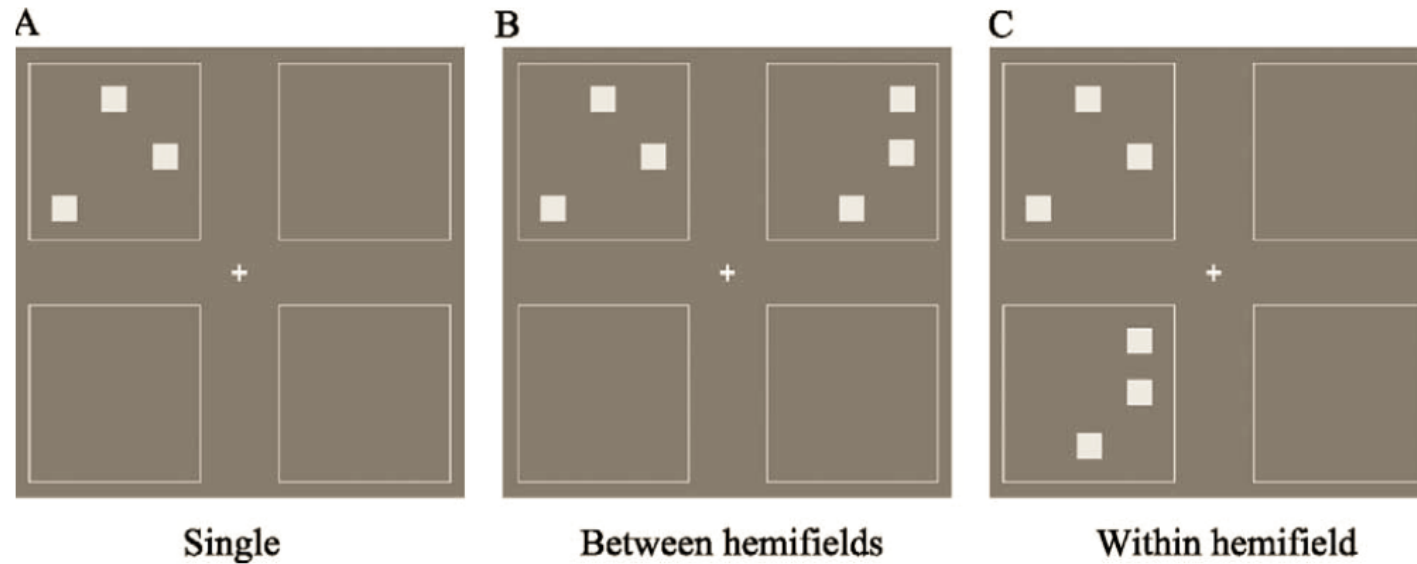
[Cabeza et al 2002 NeuroImage]

“HARATAA” | Hemispheric Asymmetry Reduction Across Tasks in Any Adult

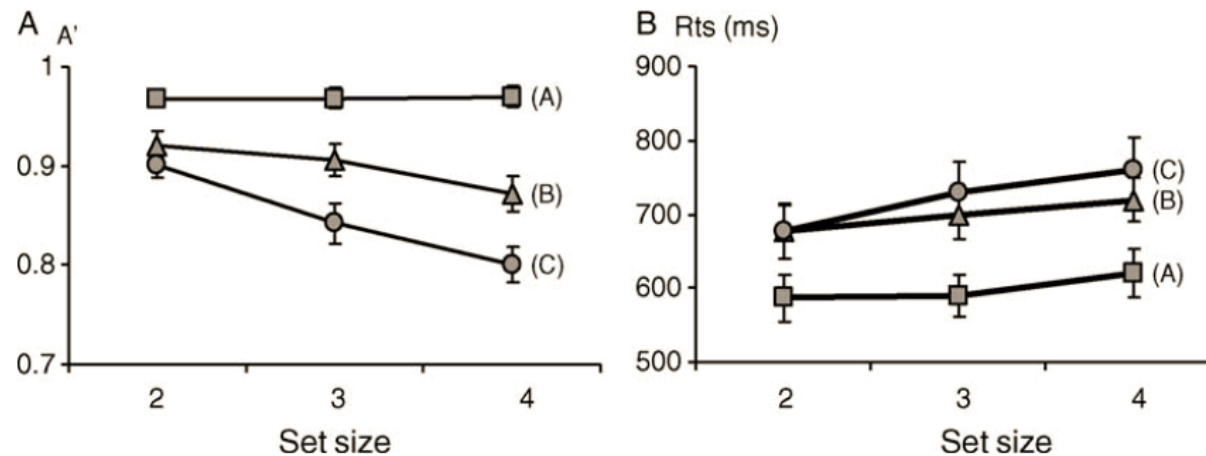


[Höller-Wallscheid et al. 2017 PNAS;
Schach et al 2023 Sci Rep]

Bilateral Field Advantage



J.-F. Delvenne / Cognition 96 (2005) B79–B88



Memory

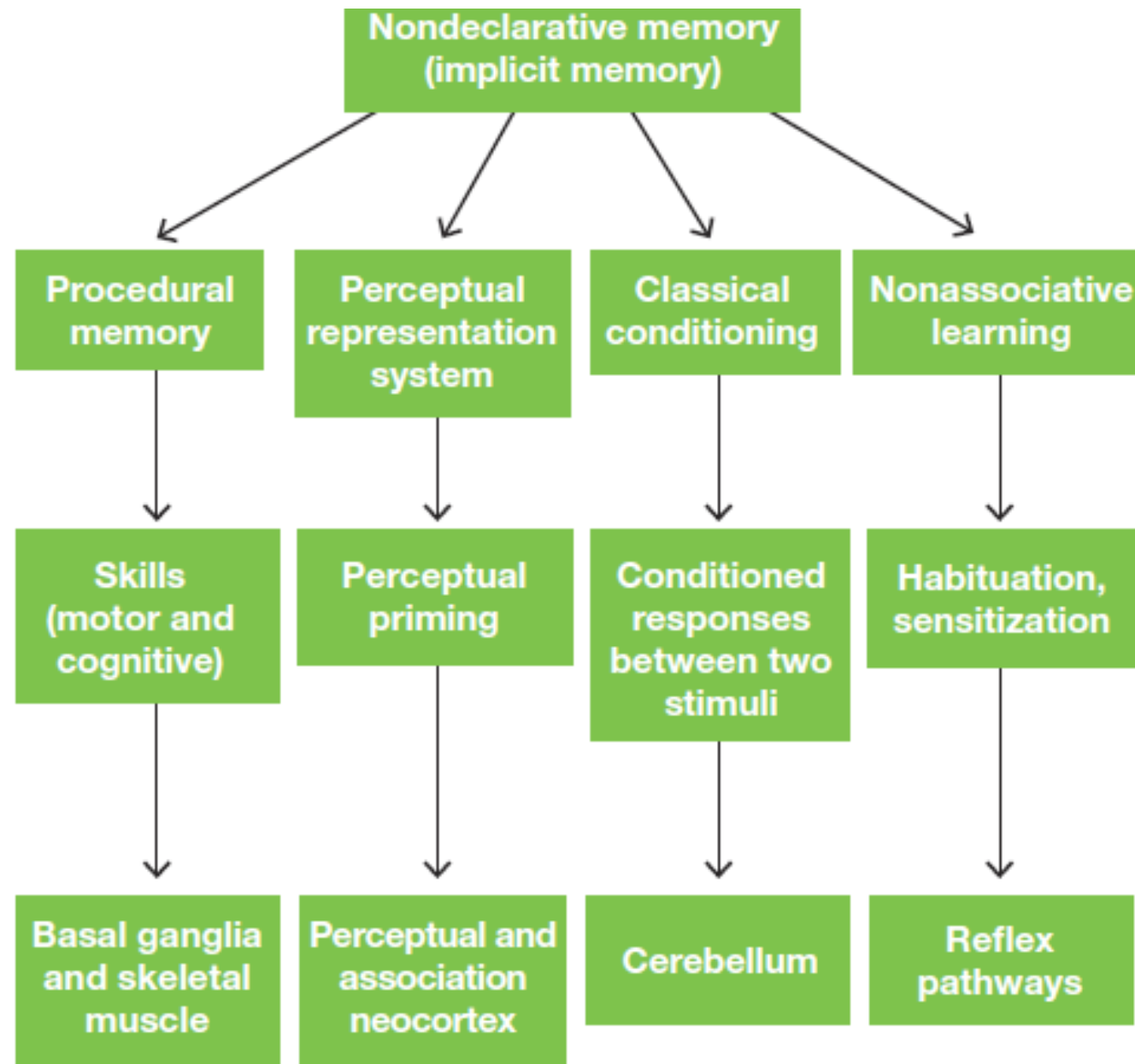
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Long-term memory ✓

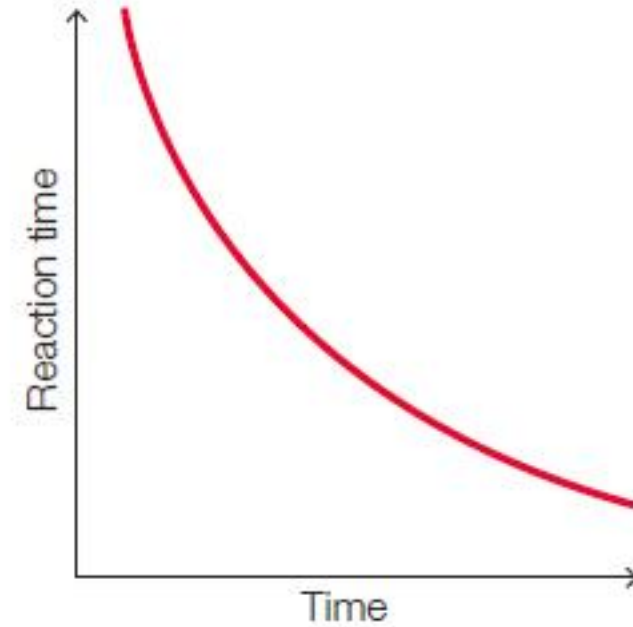
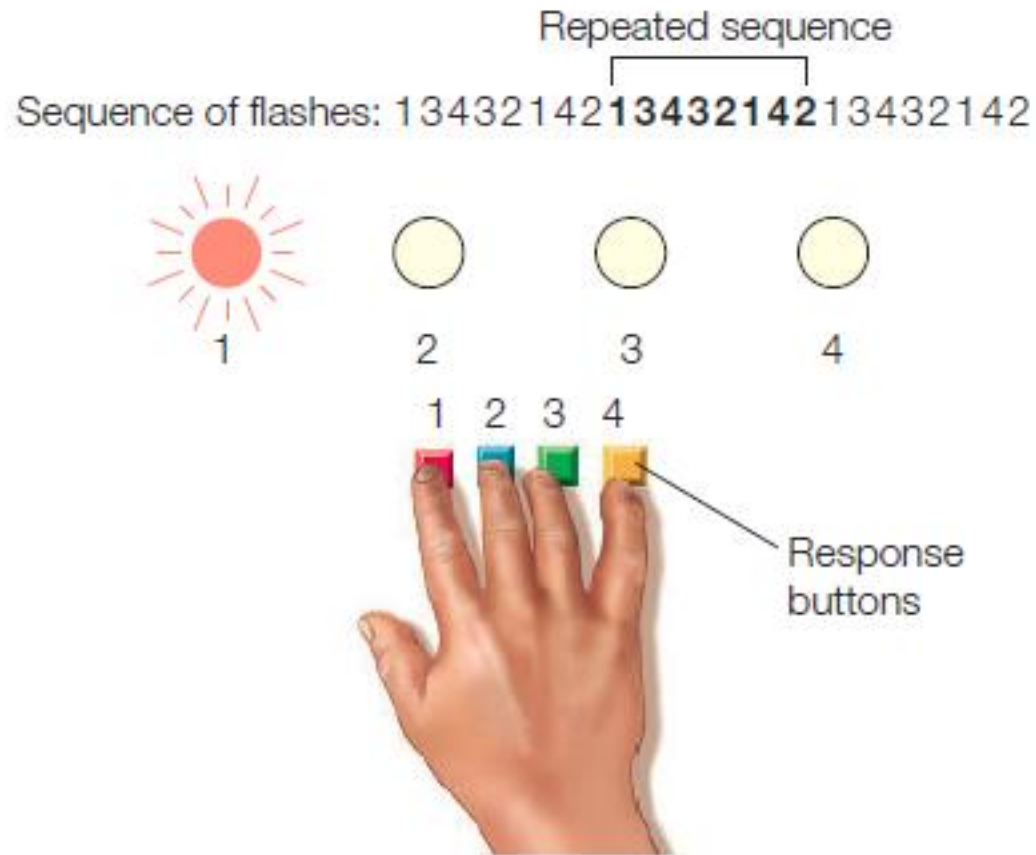
Sensory memory ✓
Short-term memory ✓
Working memory ✓

Declarative memory
(explicit memory)

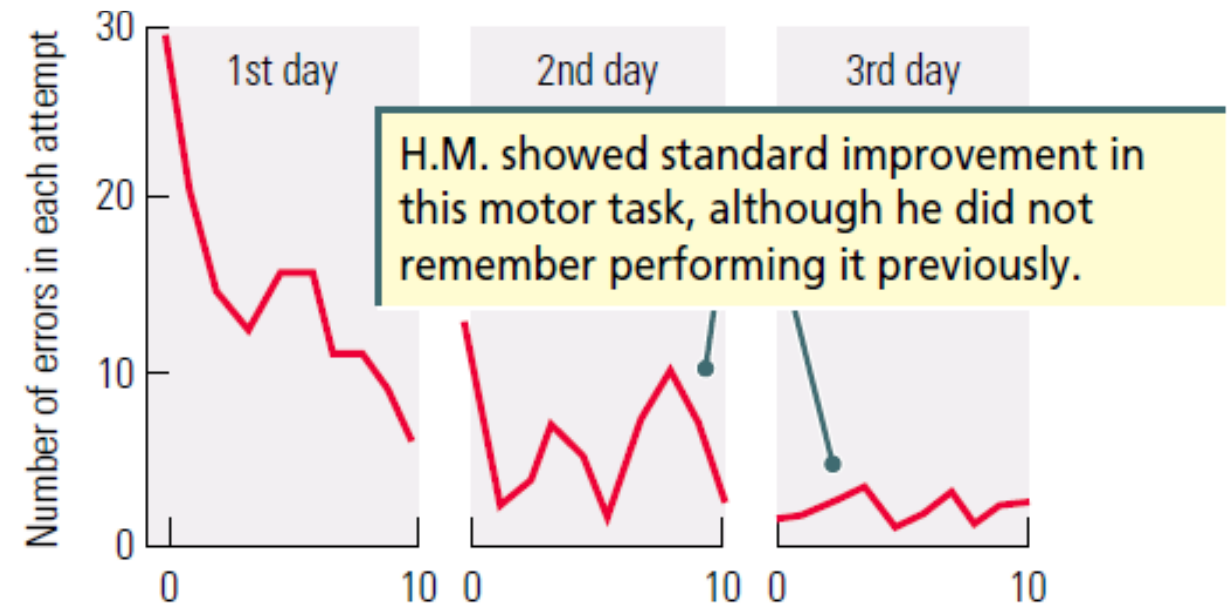
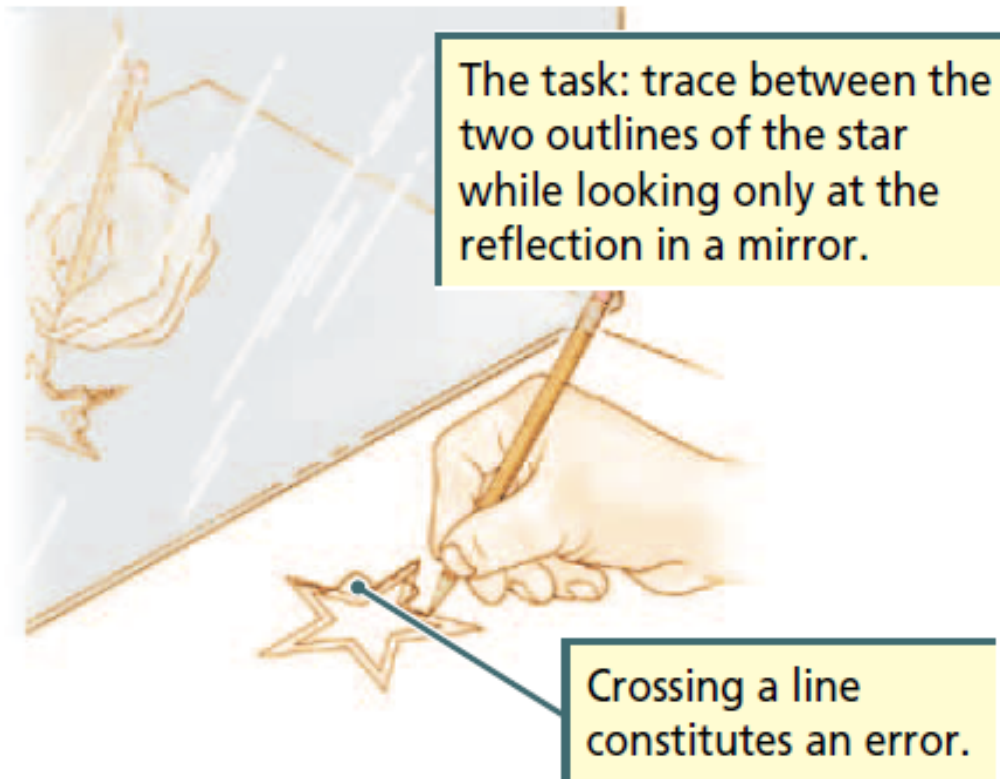
Nondeclarative memory
(implicit memory)



Procedural Memory



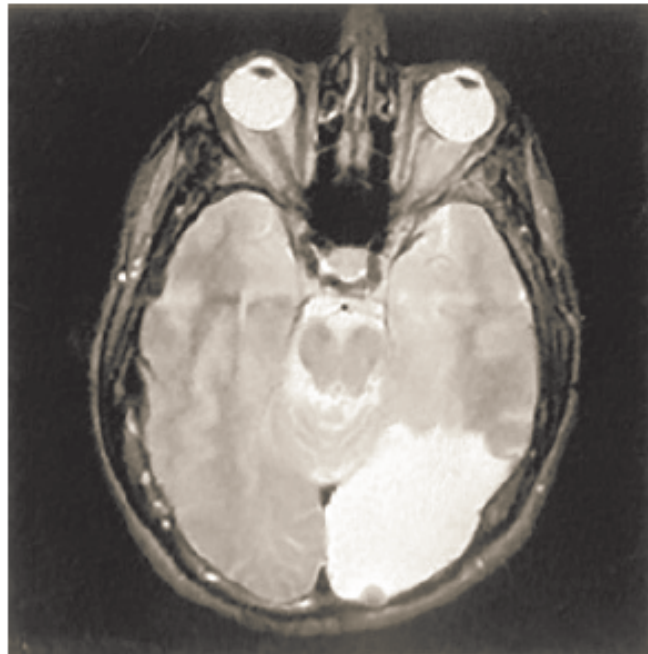
Procedural Memory in Patient H.M.



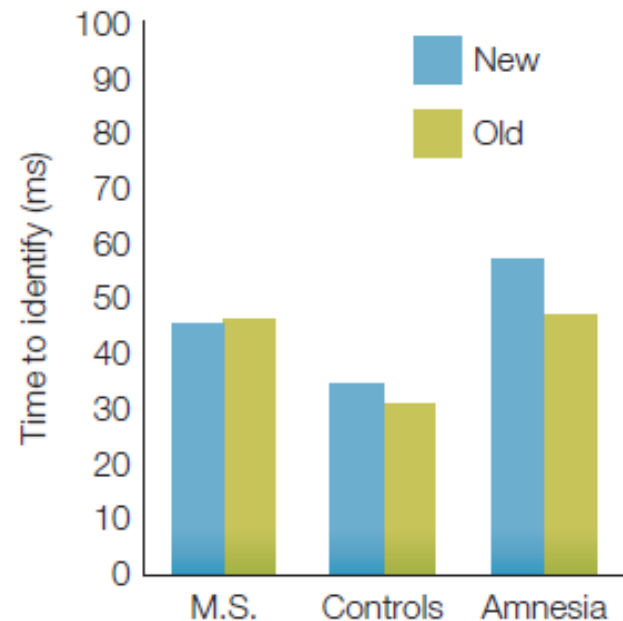
Test of Motor Memory

(A) The mirror-drawing task.
(B) Patient H.M.'s performance over three training sessions.
(Research from Milner, 1965.)

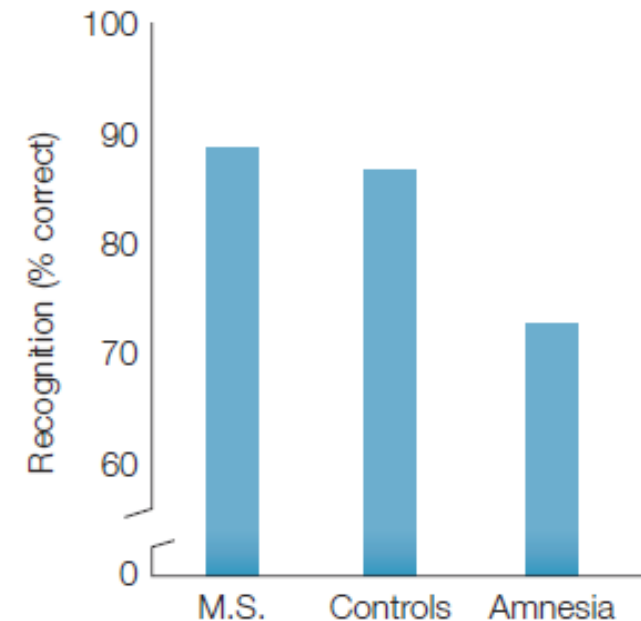
Implicit Priming vs. Explicit Memory



a



b



c

FIGURE 9.12 Double dissociation of declarative and nondeclarative memory in patient M.S.

(a) Axial MRI showing the region that was resected from M.S.'s right occipital area. (b) The average time it took to identify words in the implicit memory test. "Old" words were previously presented words. M.S. showed no priming effect, whereas both controls and participants with amnesia showed a timing difference between the new and old words. (c) Accuracy of the recognition test of explicit memory. M.S. had no problem recognizing words in this test of explicit memory, in contrast to patients with anterograde amnesia.

[John Gabrieli Lab, Stanford, 1995]

Classical Conditioning

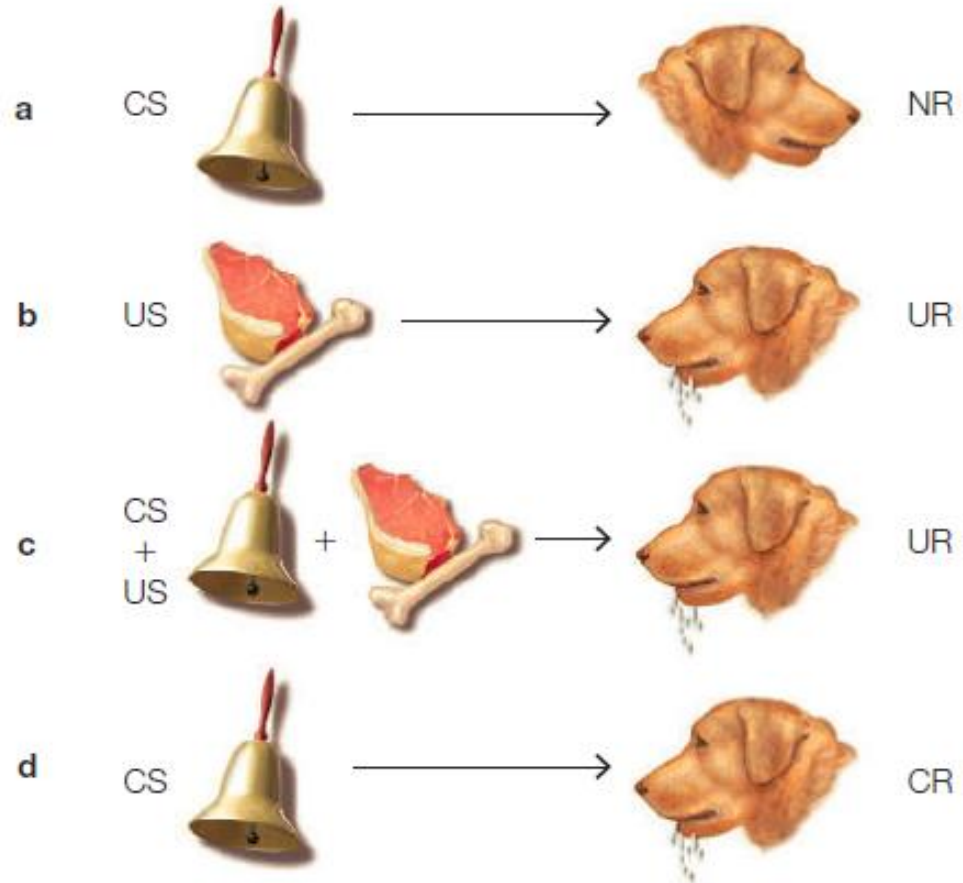
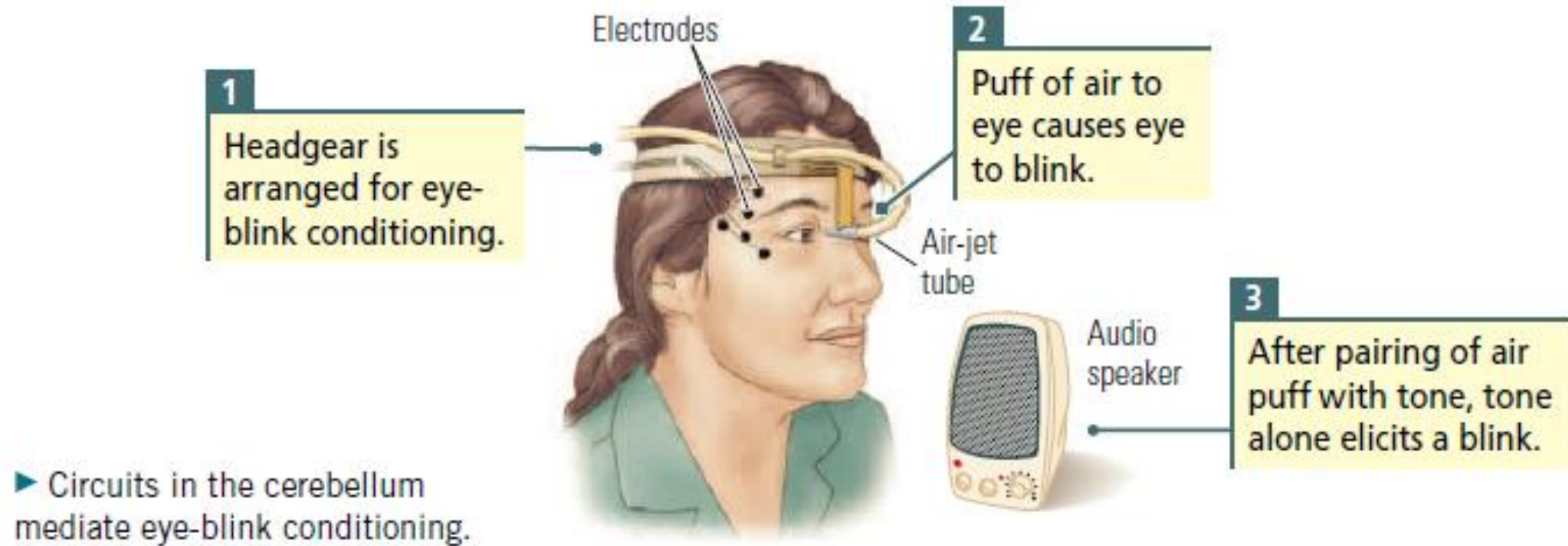
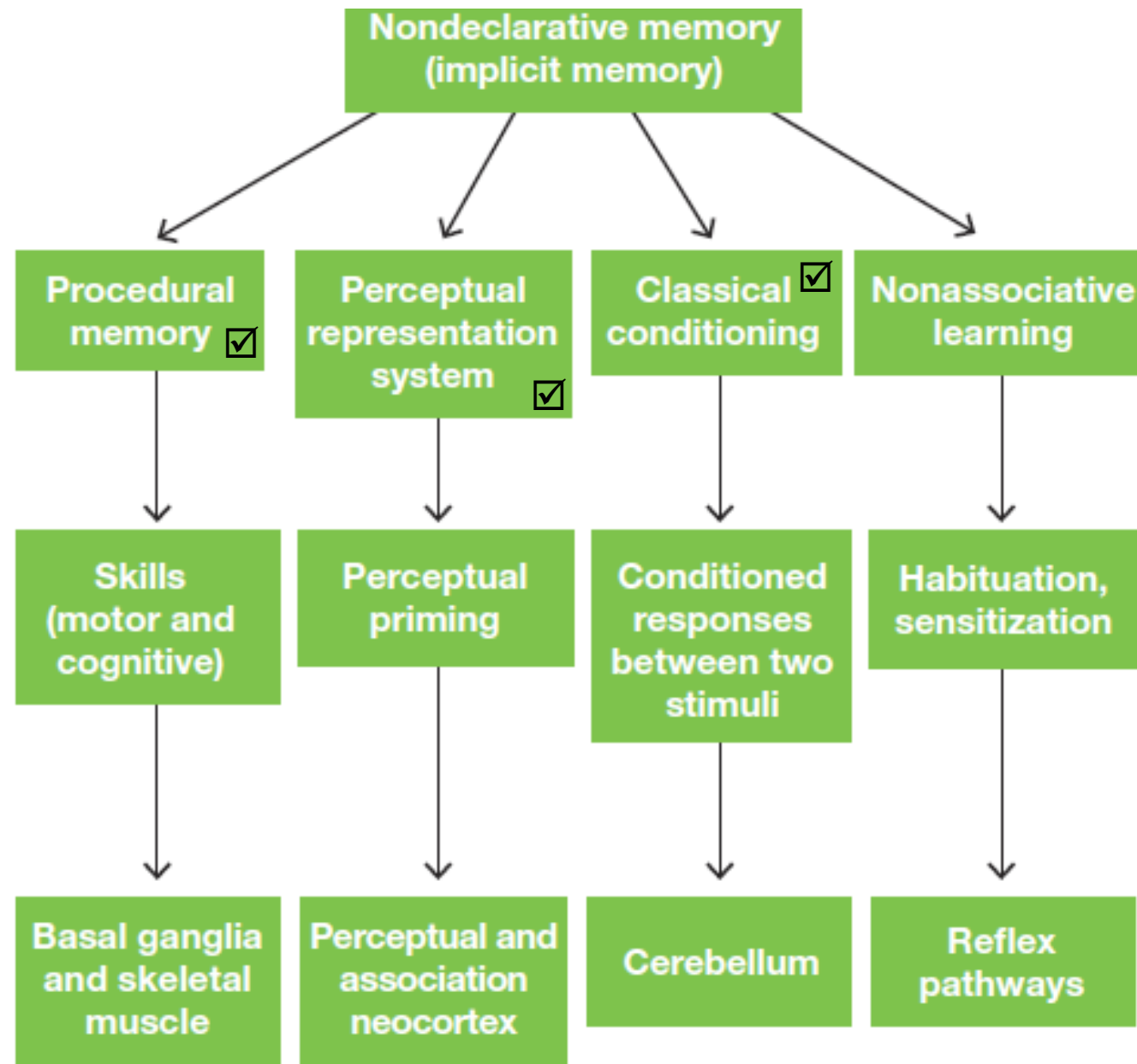


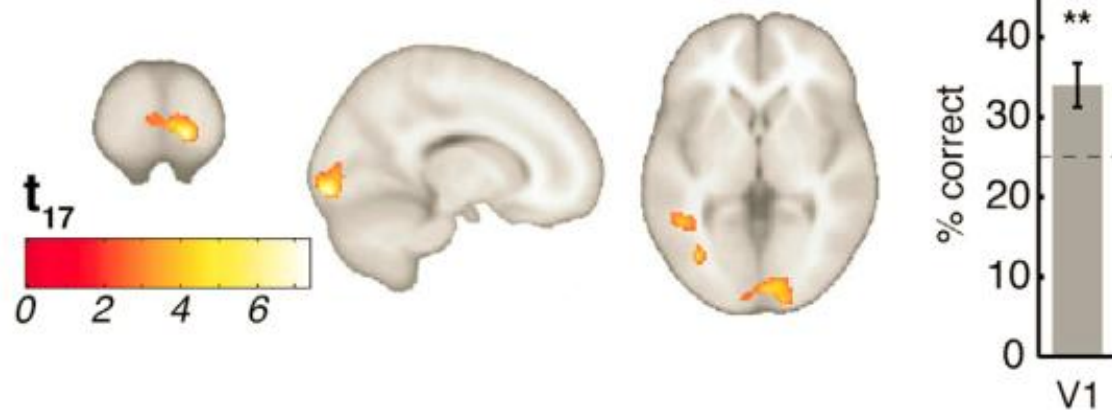
FIGURE 9.13 Classical (Pavlovian) conditioning.

Classical Conditioning

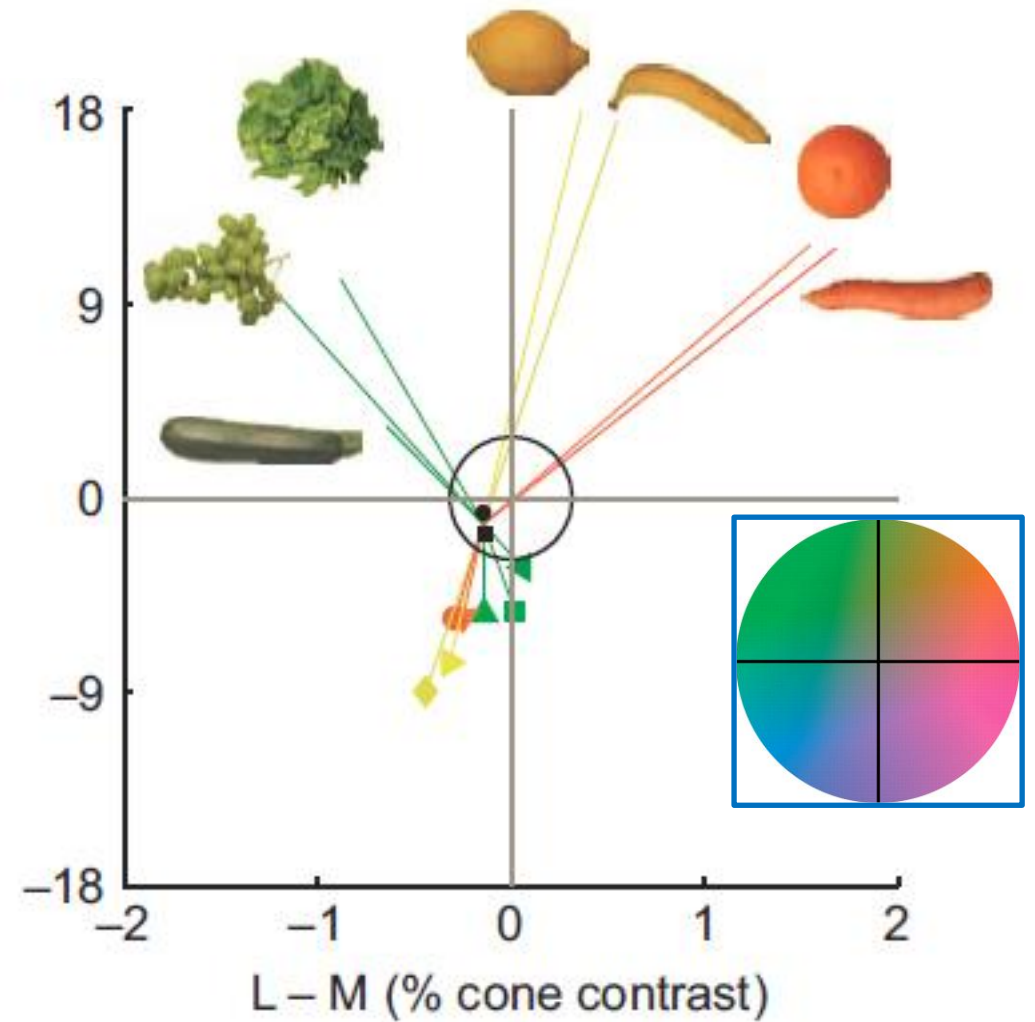




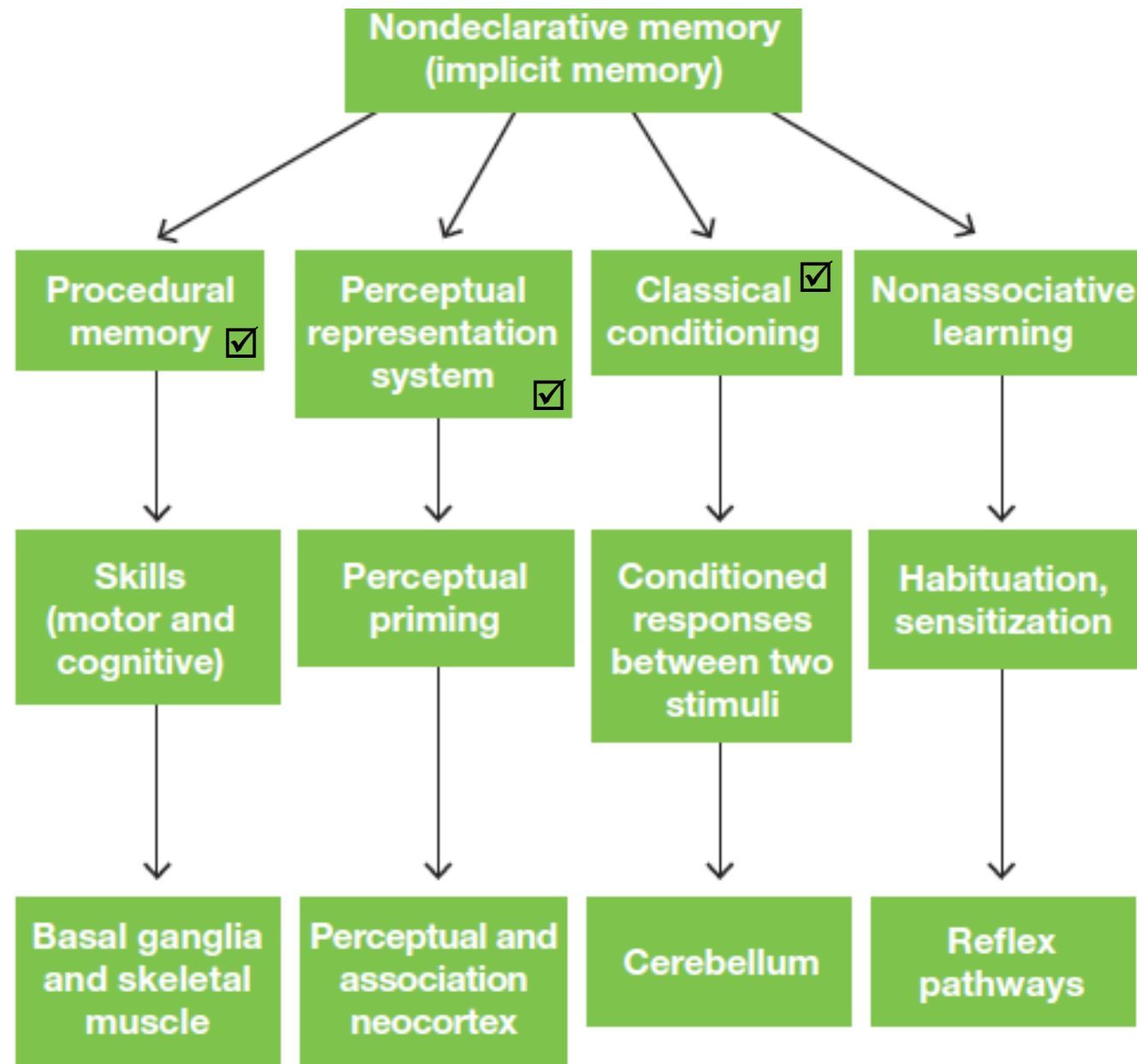
Memory Colors



Bannert & Bartels 2013 Curr Biol



Hansen et al., 2006 Nat Neurosci



Hollow Face Illusion



“Free Energy”

“Predictive Coding”

“Internal Models”

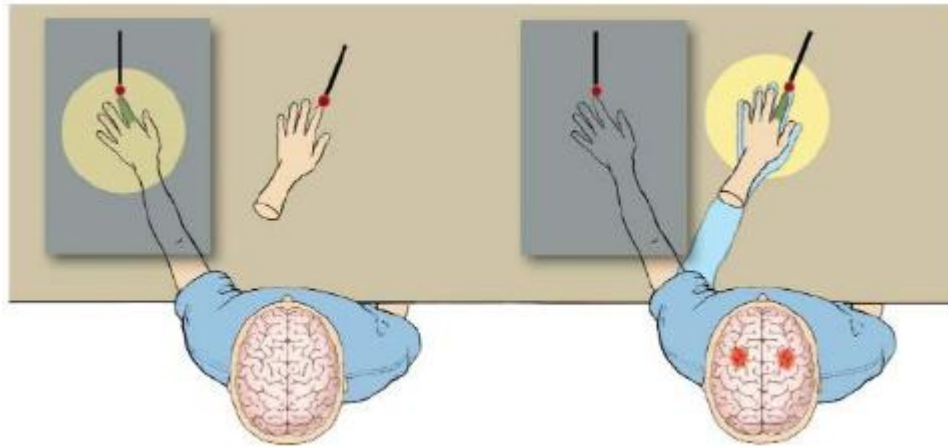
“Free Energy”

“Predictive Coding”

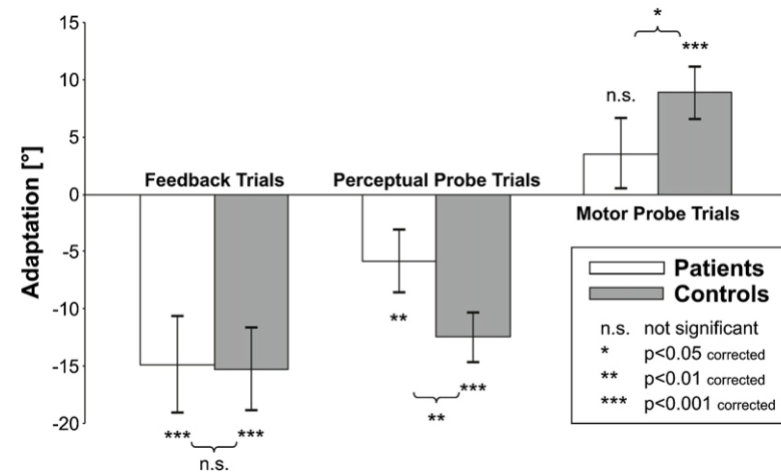
“Internal Models”

... and the “Minimal Self”

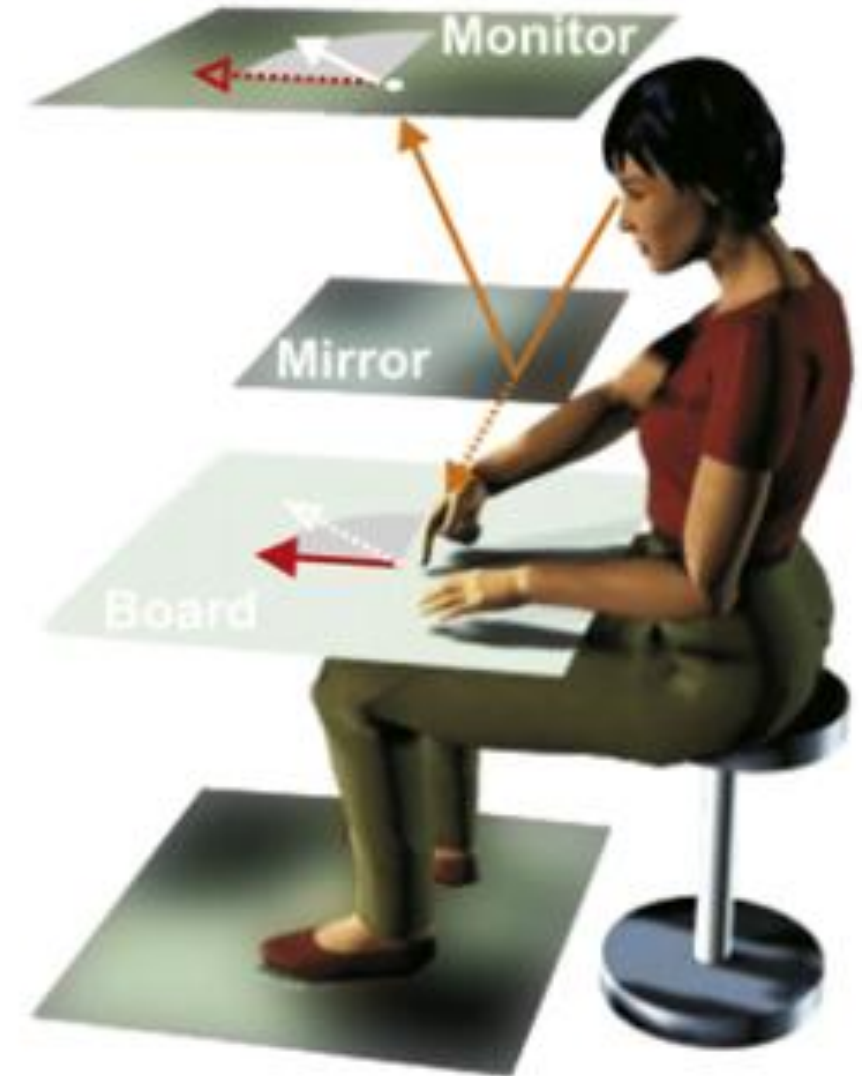
„Ownership“ – the Rubberhand Illusion



„Agency“



Synofzik et al., 2008 Curr Biol



... and the “Minimal Self”

Memory

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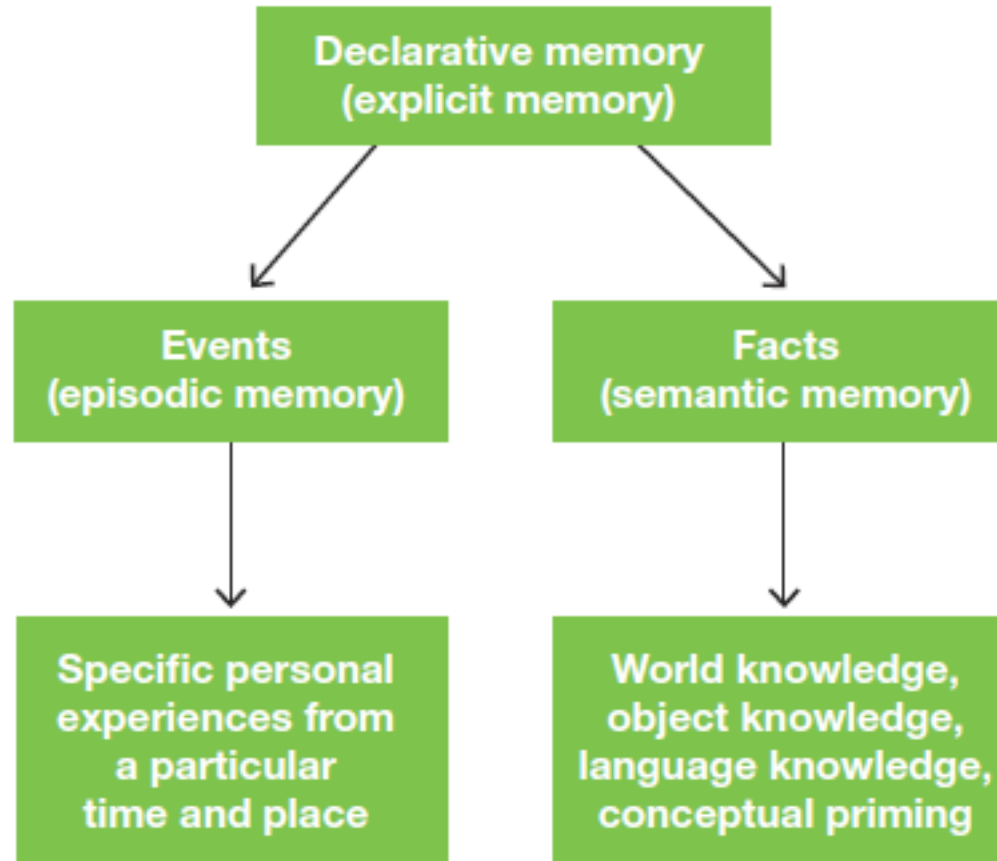
Long-term memory ✓

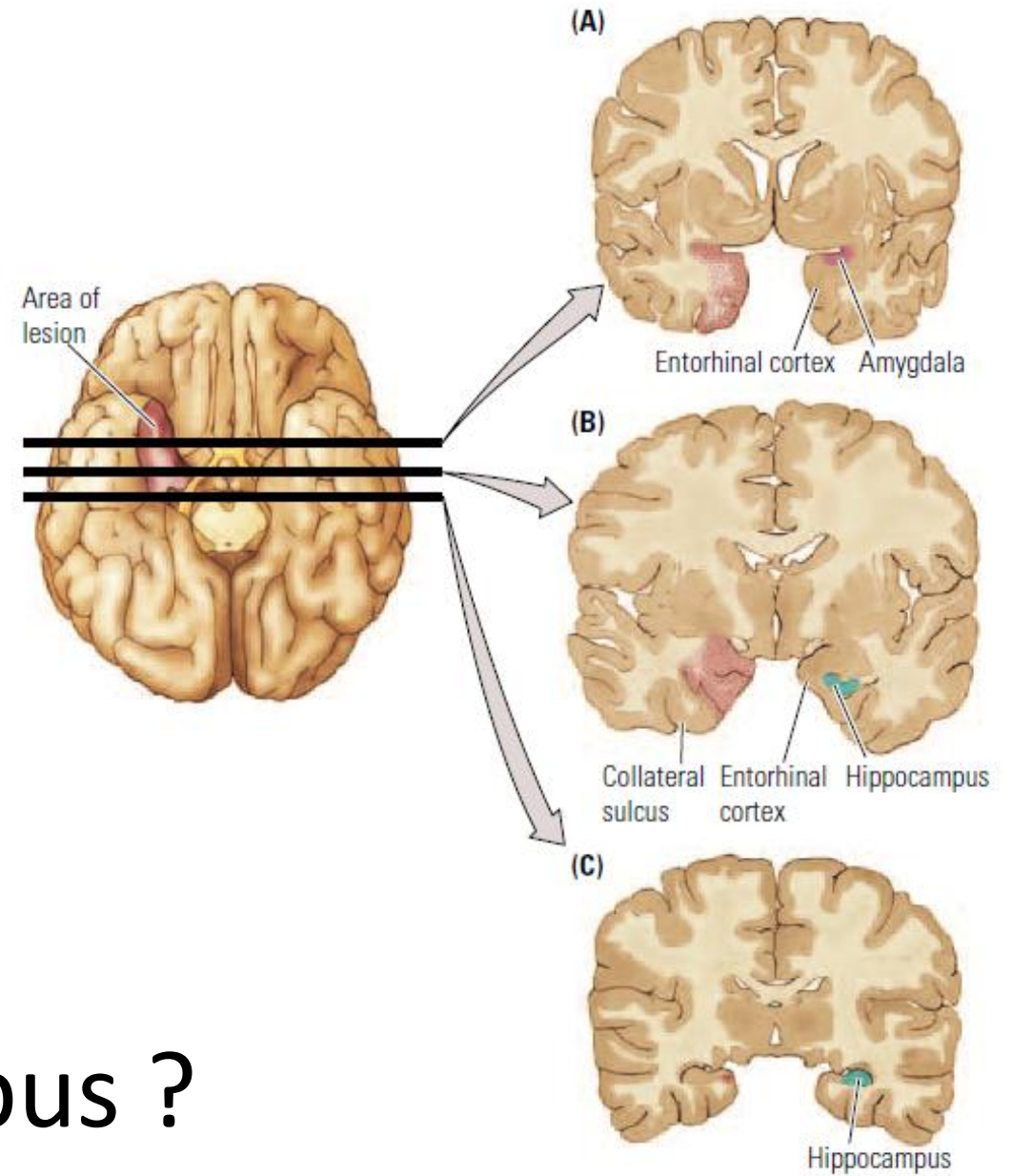
Sensory memory ✓
Short-term memory ✓
Working memory ✓

Declarative memory
(explicit memory)

Nondeclarative memory
(implicit memory) ✓

Declarative Long Term Memory: Events vs. Facts





The role of the Hippocampus ?

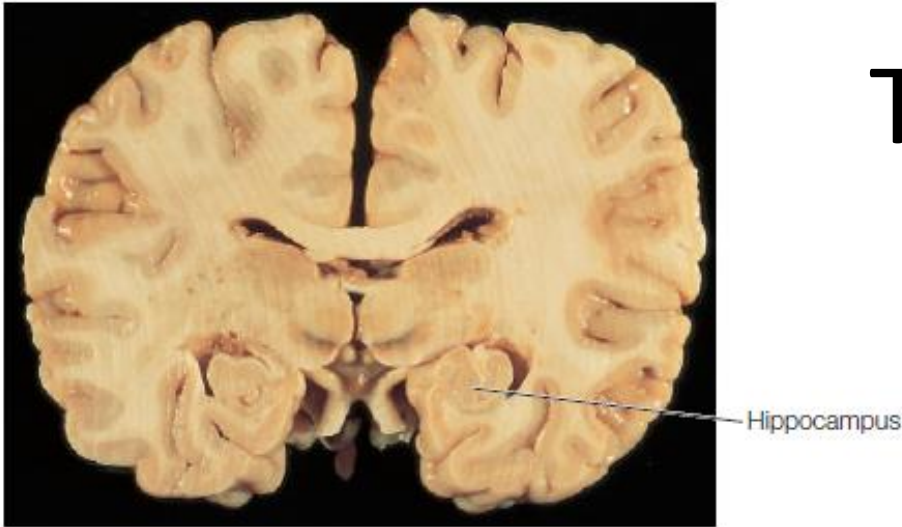
The role of the Hippocampus ?

Patient R.B.

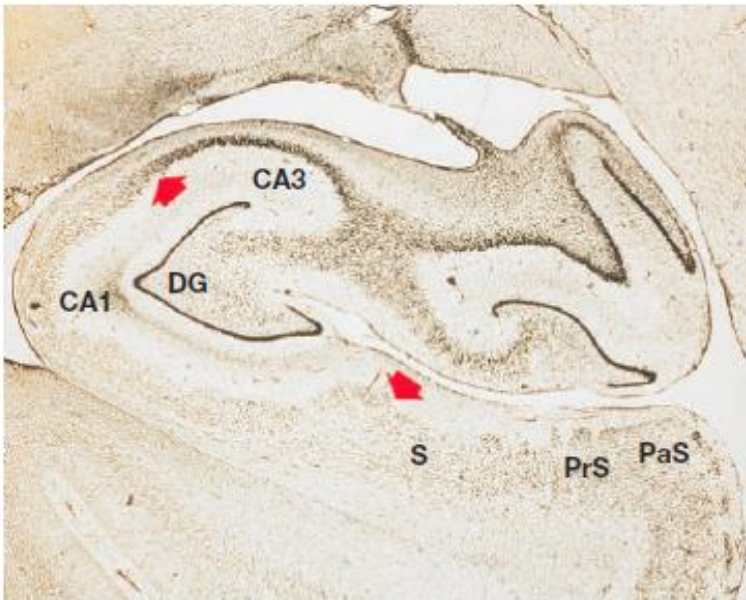
Lesion restricted to CA1 Pyramidal Cells due to ischemic episode during heart bypass surgery

Anterograde Amnesia similar to H.M.
Retrograde Amnesia (1-2 years)

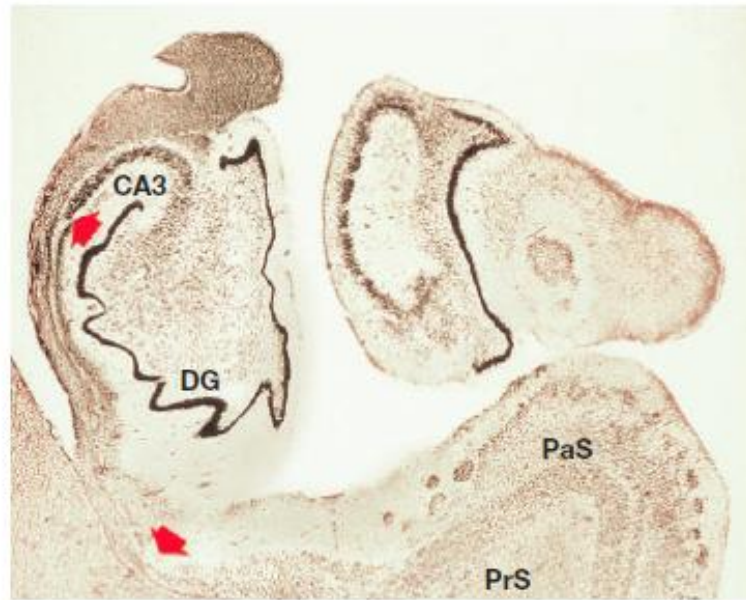
Reported by Zola-Morgan et al., (1986)



a

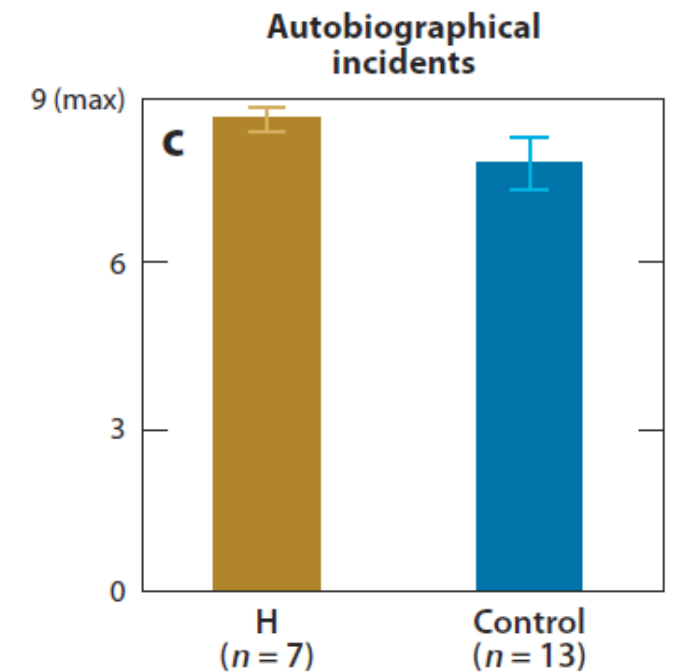
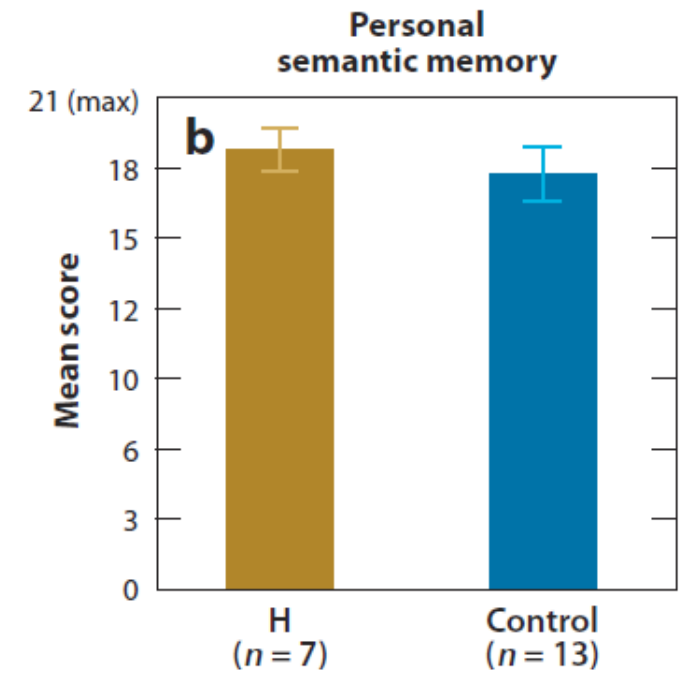
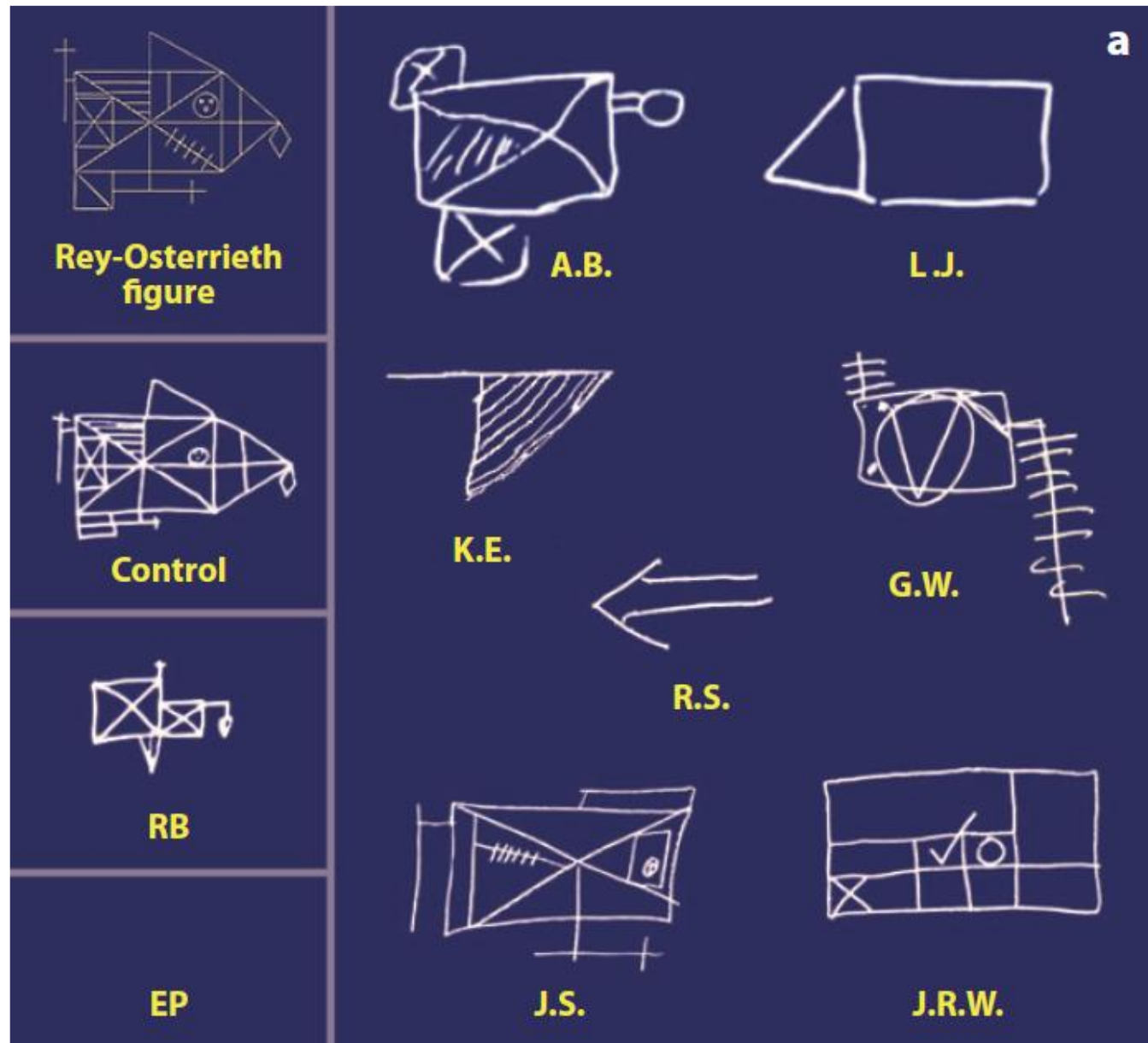


b Nonamnesic



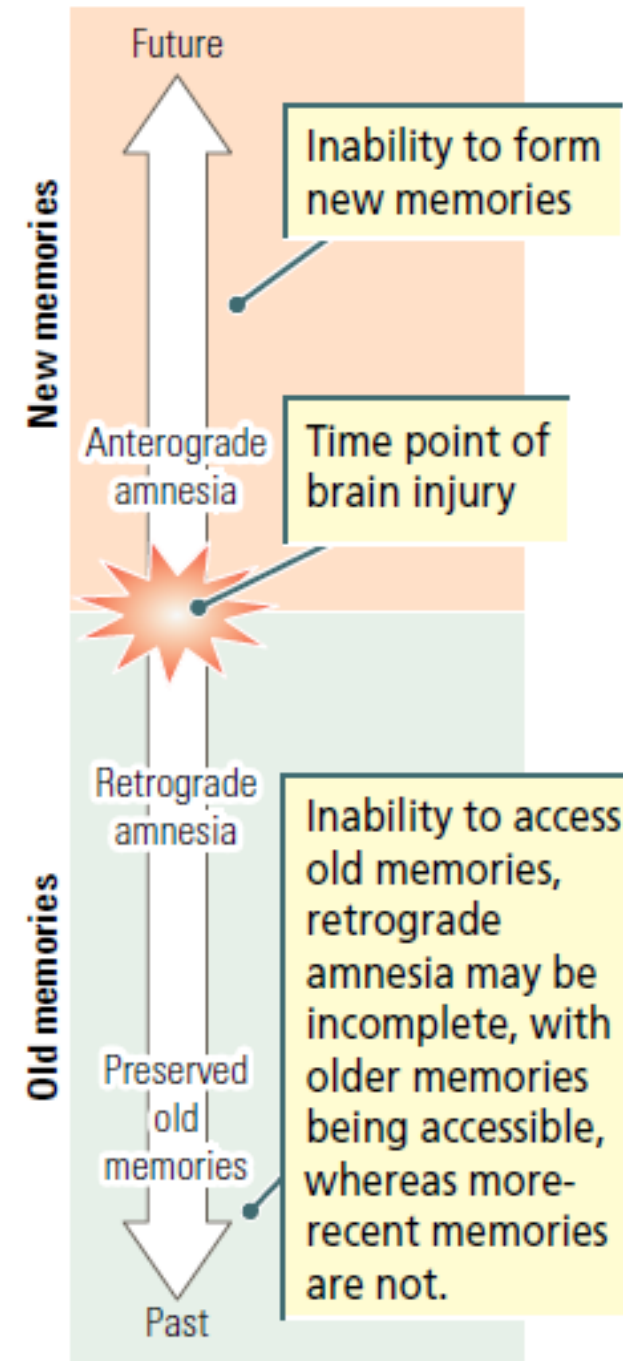
c Case R.B.

FIGURE 9.17 Comparison of R.B.'s brain with that of a nonamnesic participant.



Amnesia

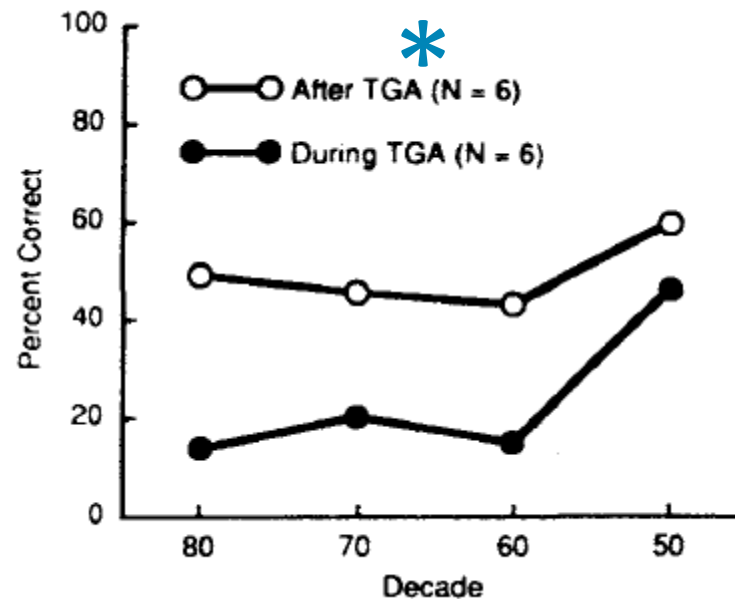
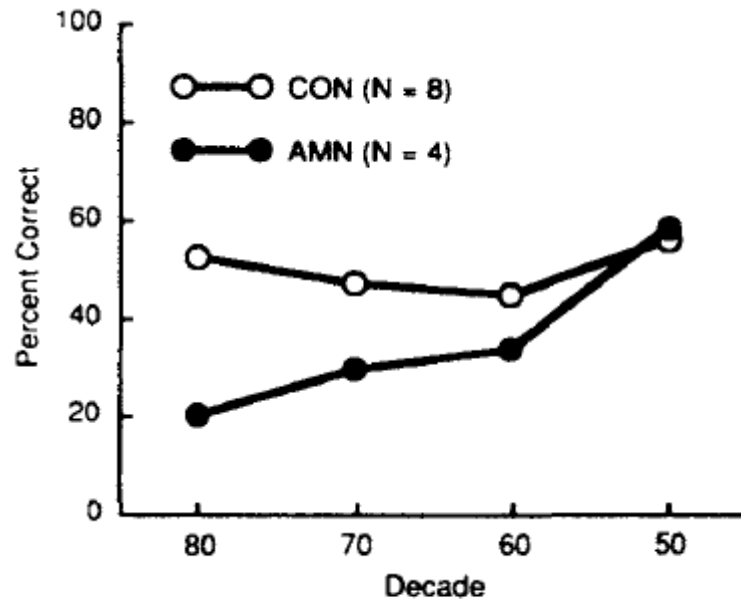
Anterograde vs. retrograde amnesia



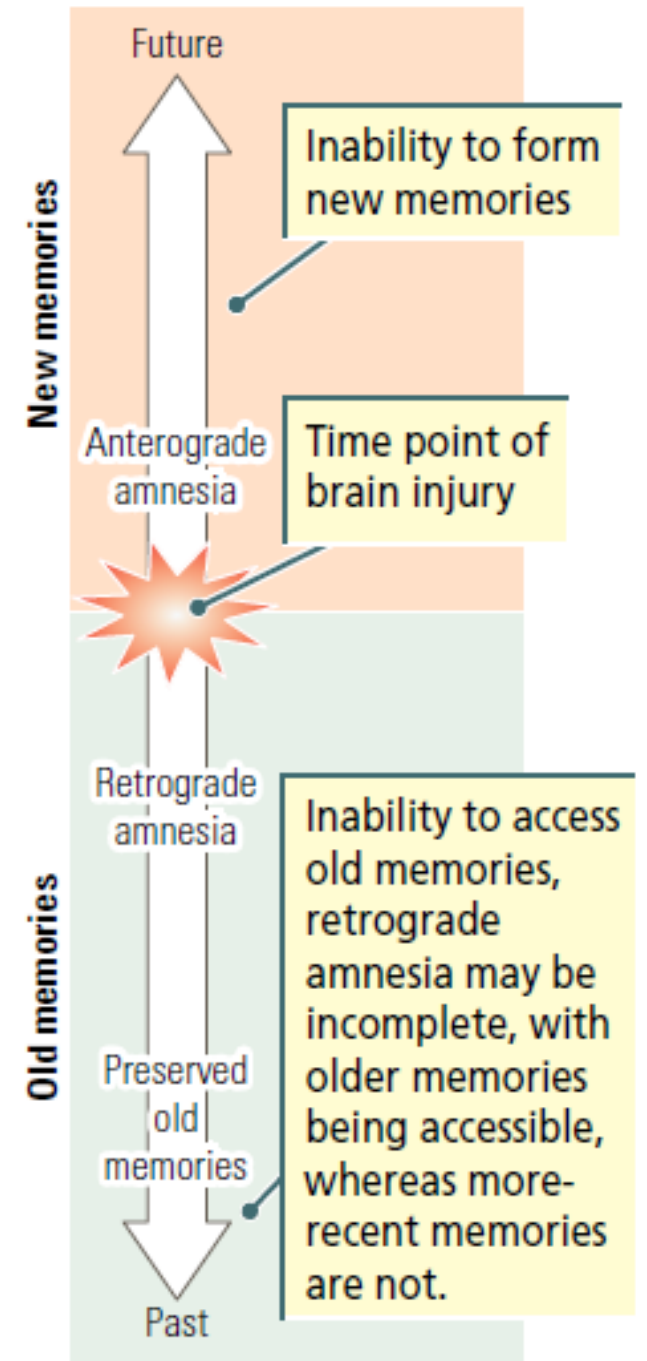
Retrograde Amnesia

Ribot's Law (1882): "the new perishes before the old"

Recall Rates: Public Events-Test

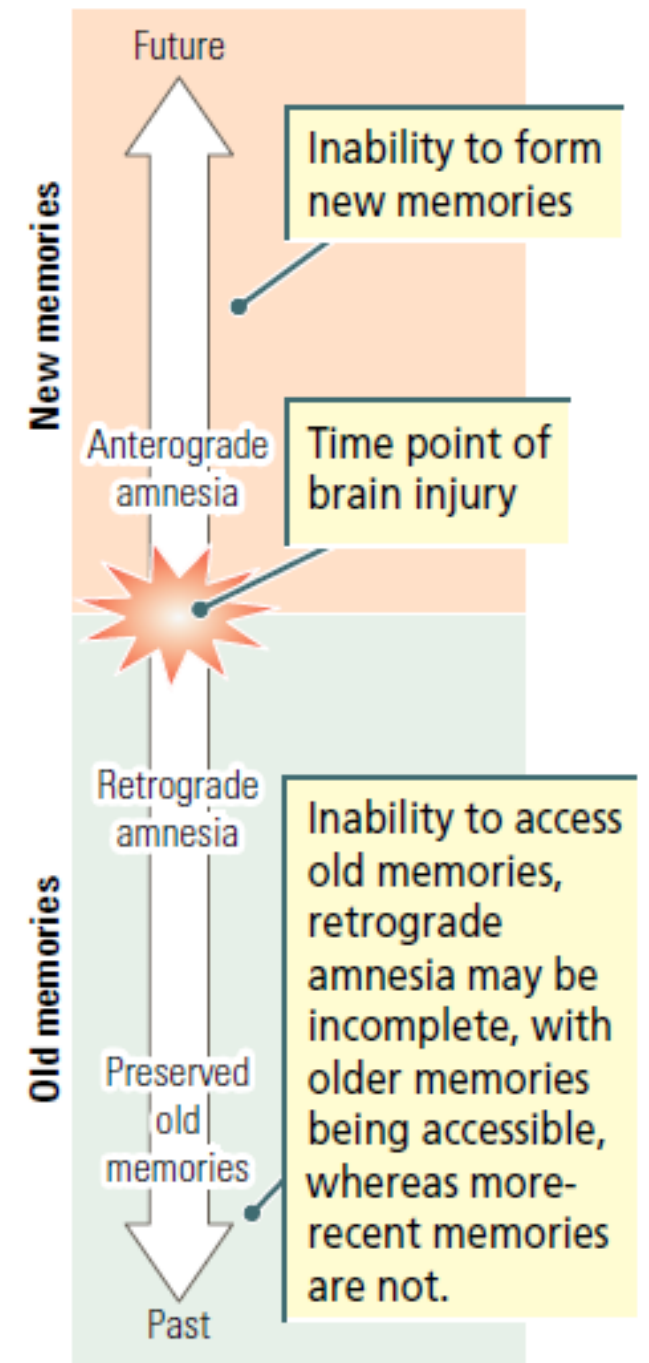


* Transient Global Amnesia



- * Further evidence that the hippocampus is involved in long-term memory acquisition comes from patients with **transient global amnesia (TGA)**. This syndrome has a number of causes, but it is triggered most commonly by physical exertion in men over 50 and by emotional stress in women over 50. In this situation, the typical blood flow is disrupted in the brain. In particular, the **vertebrobasilar artery system**, which supplies blood to the medial temporal lobe and the diencephalon, has been implicated as a critical site. The result is a transient ischemia that later returns to normal.

The role of the Hippocampus?

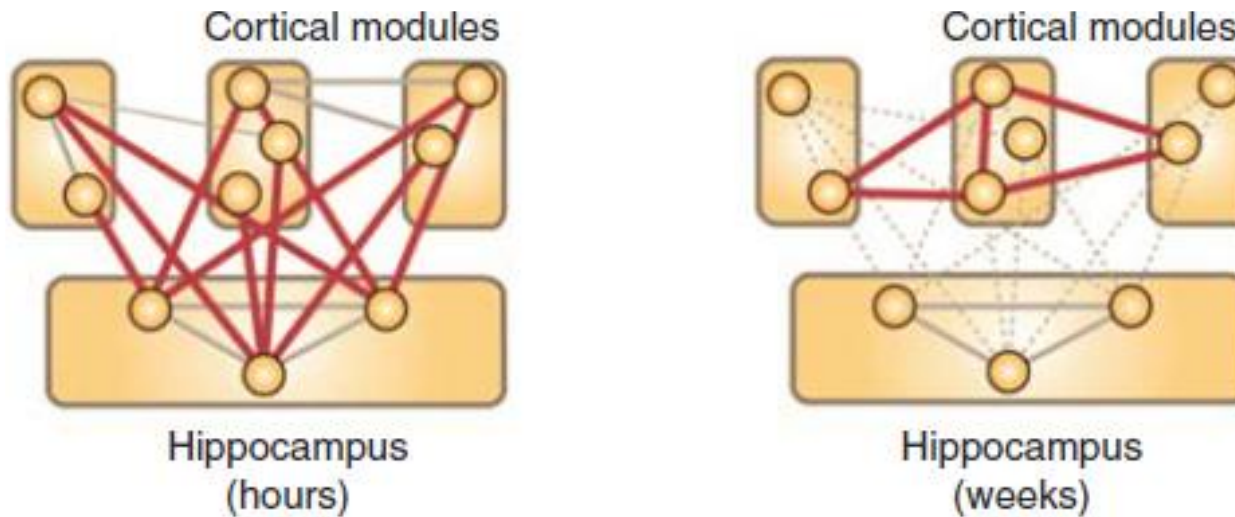


The role of the Hippocampus?

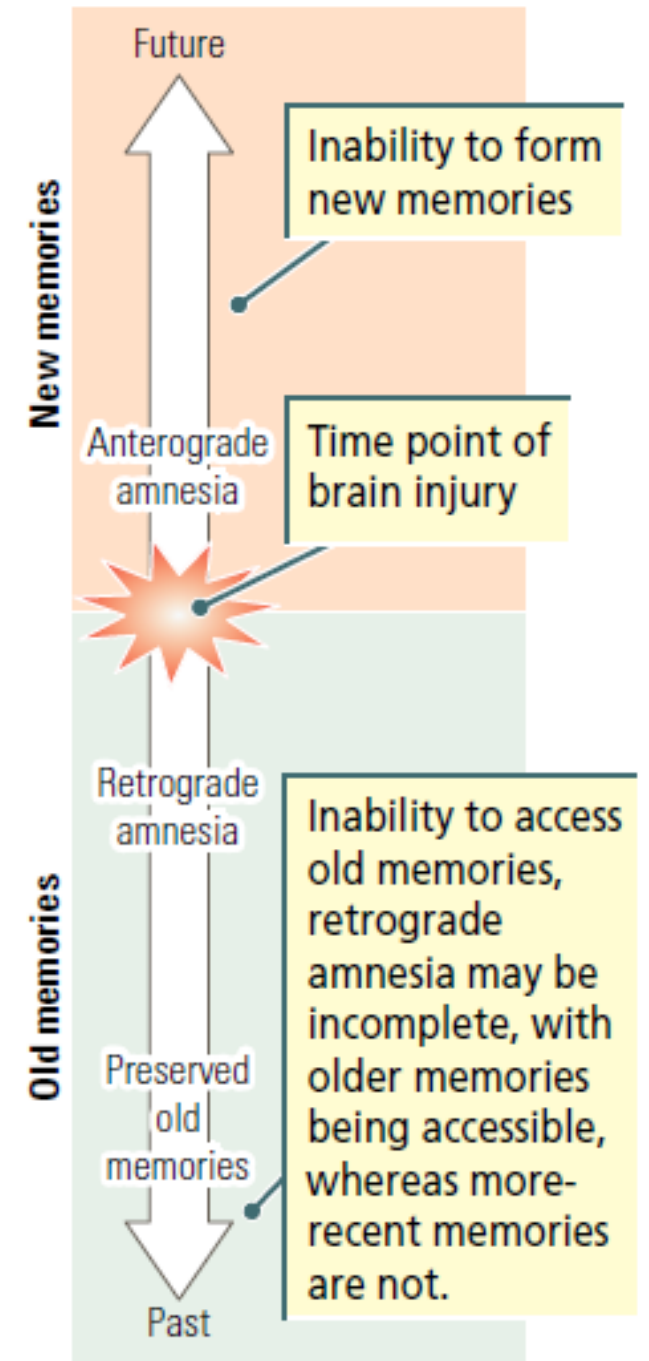
... episodic memory consolidation!

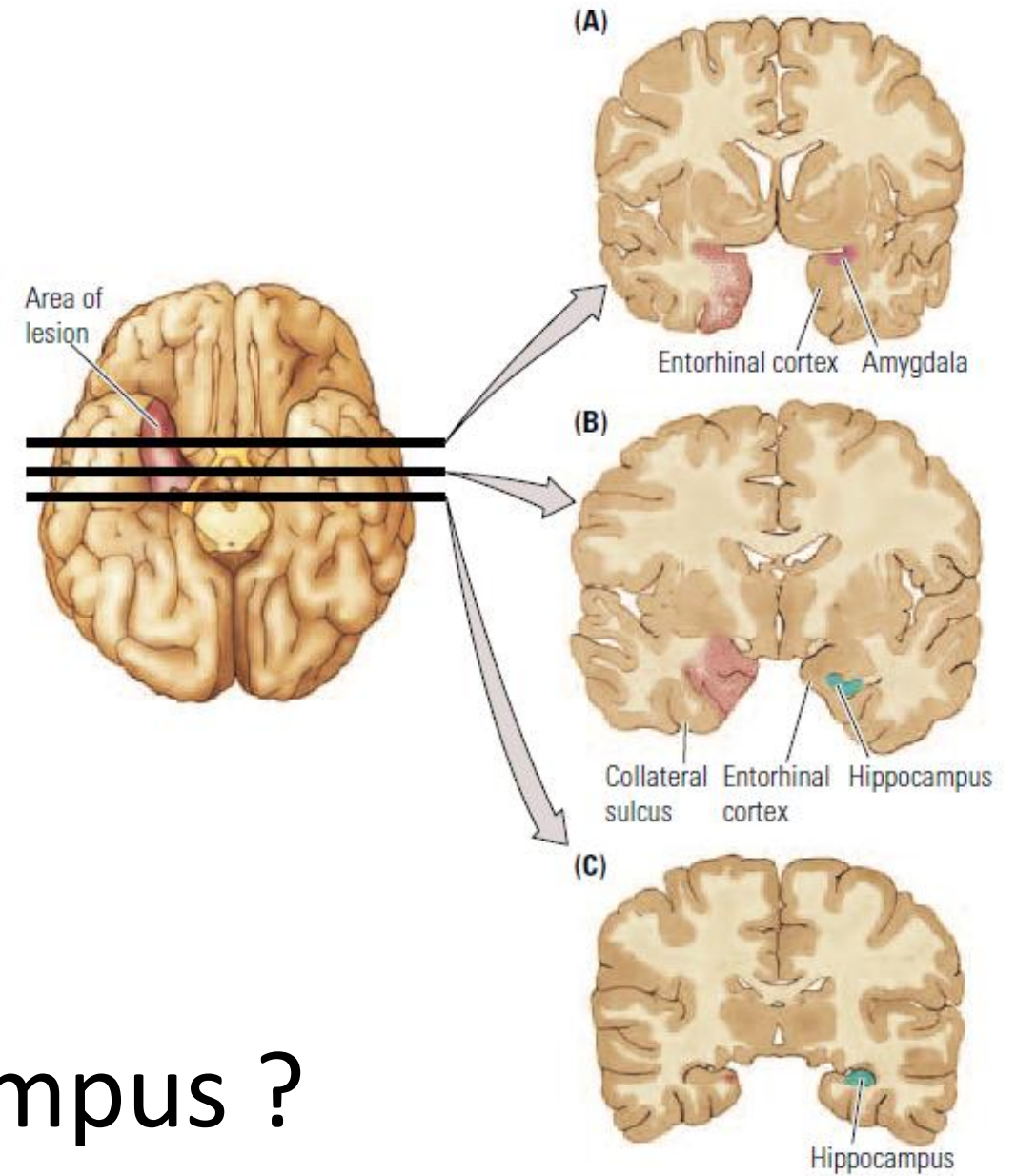
Compare:

- Standard Consolidation Theory (e.g. Squire et al. 2015)



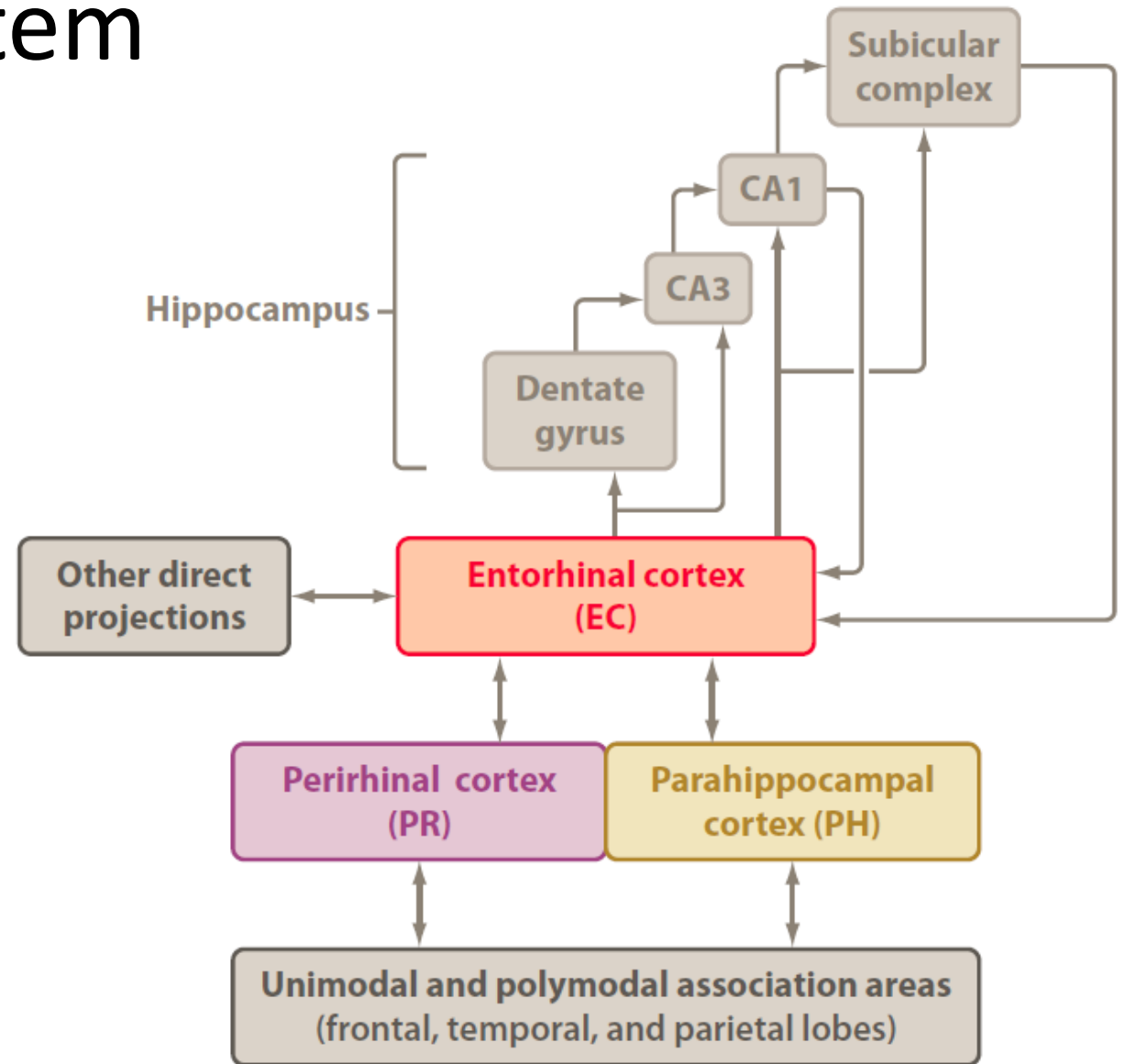
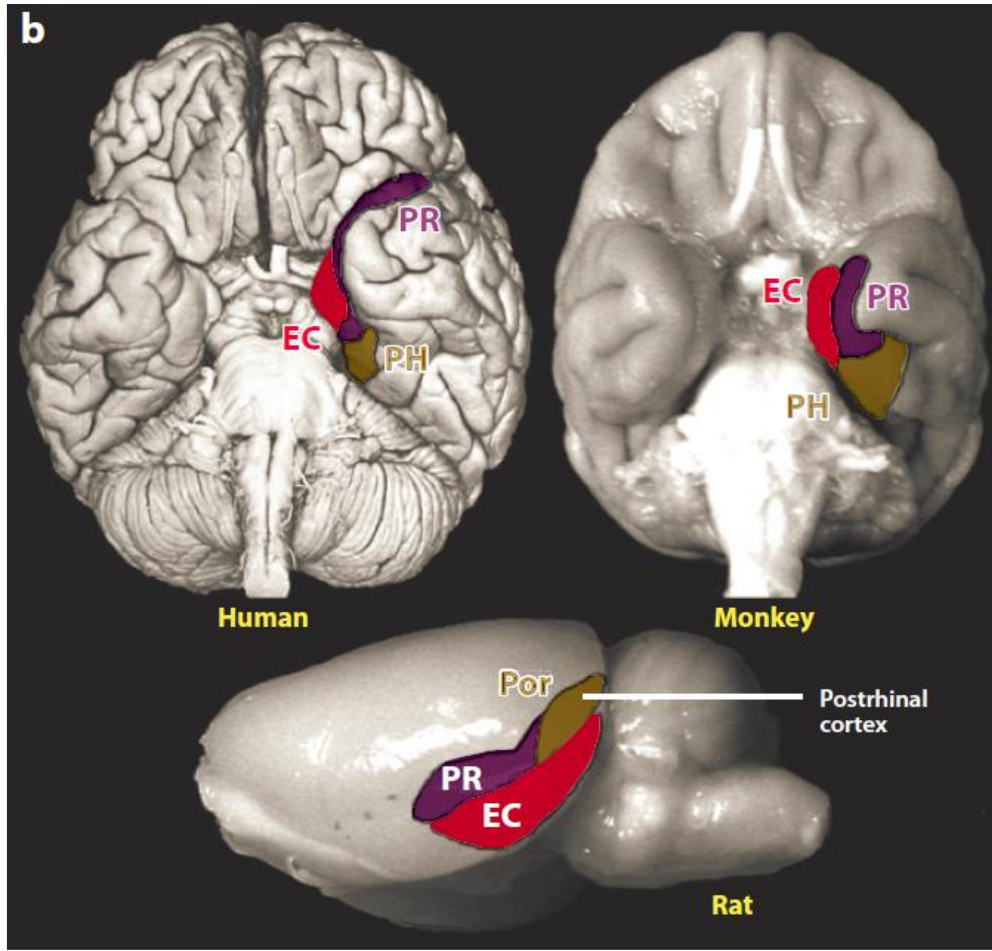
- Multiple Trace Theory (Nadel & Moscovitch, 1997)

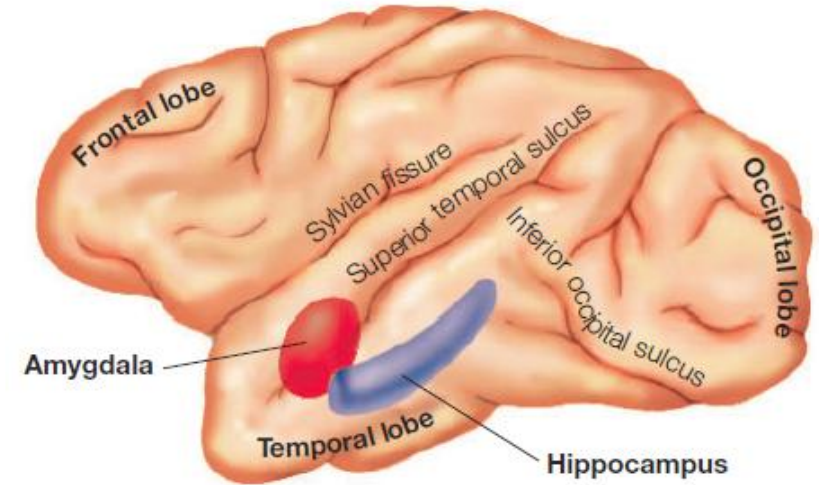
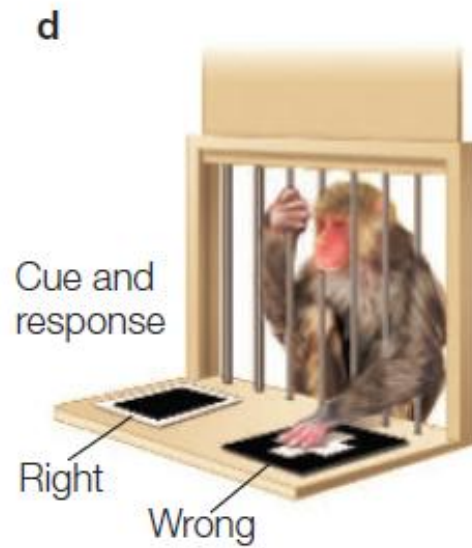
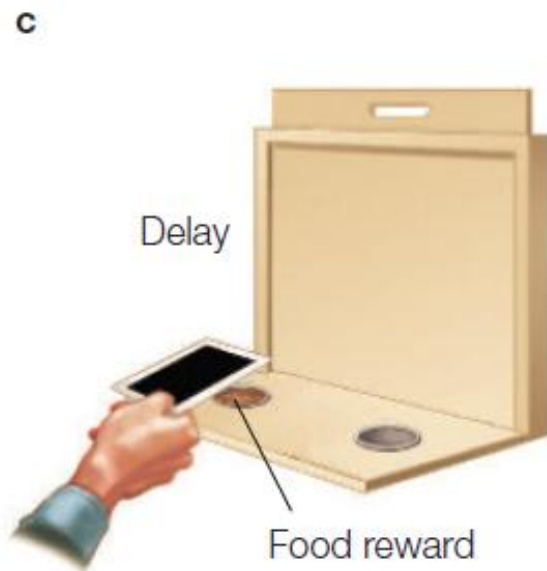
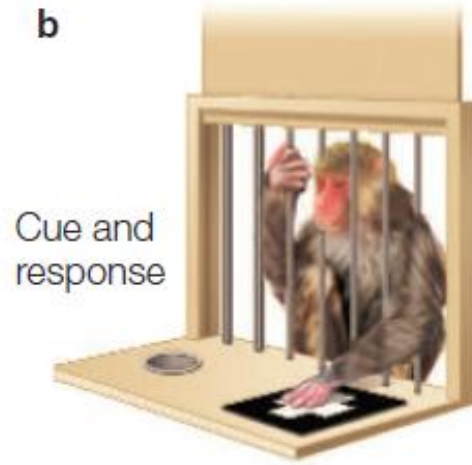
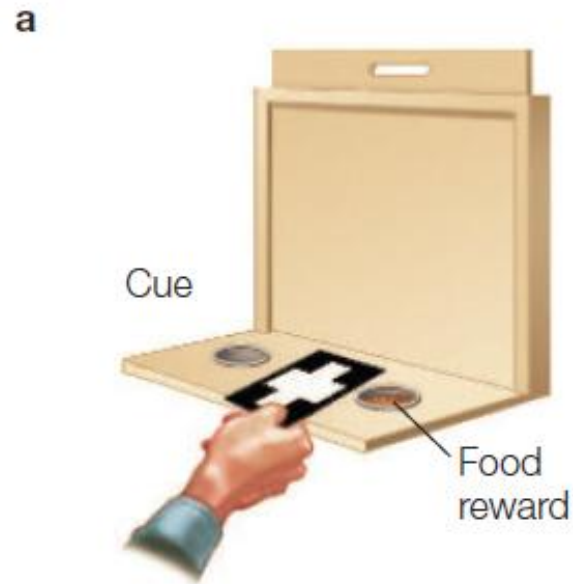




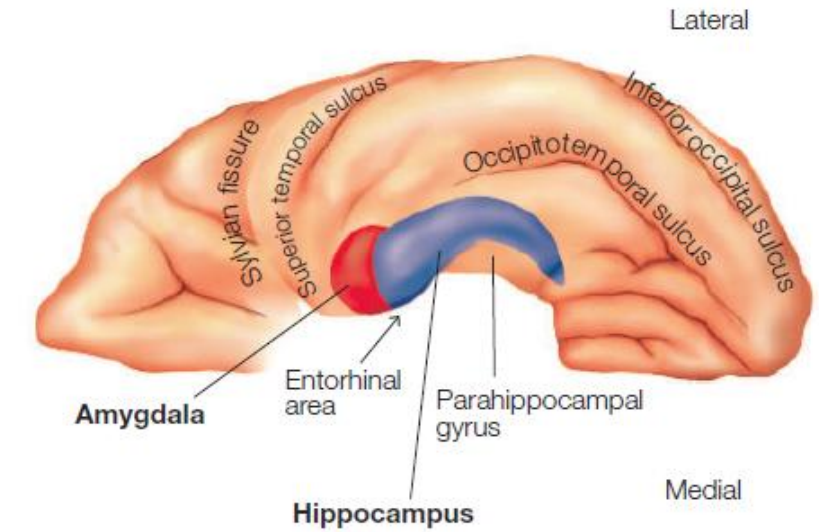
A sole role of the Hippocampus ?

The MTL Memory System



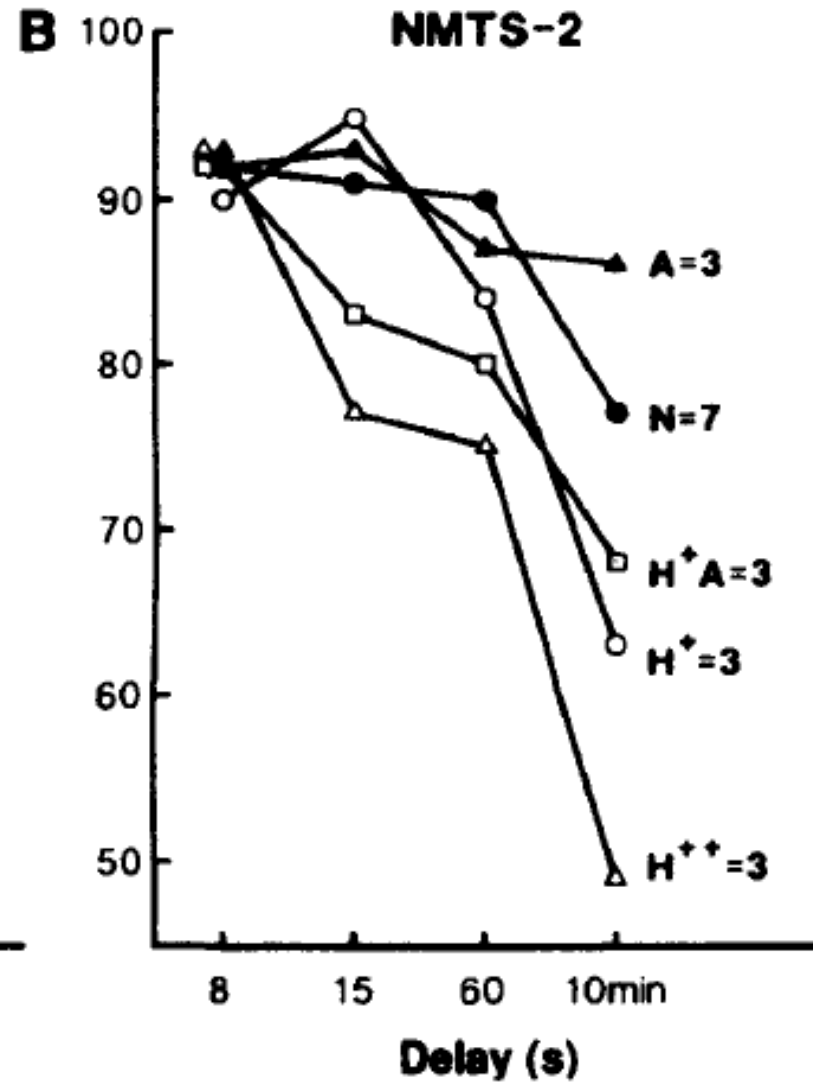
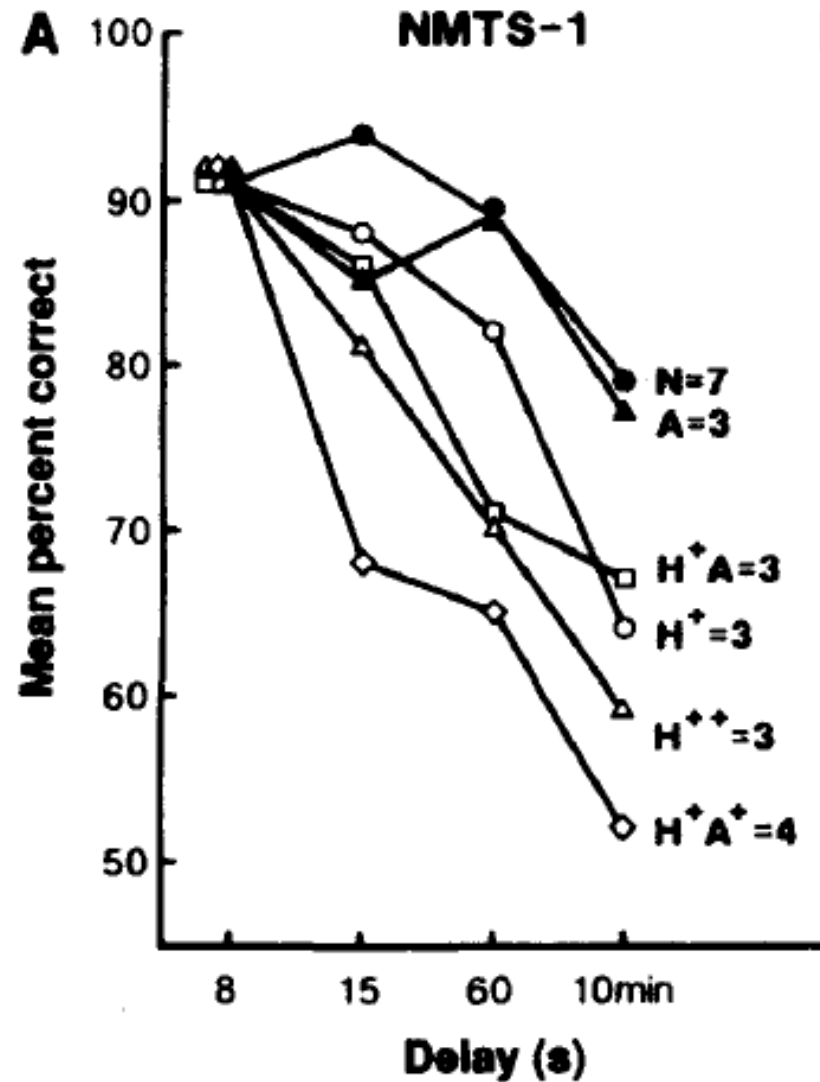


a Lateral view



b Ventral view

Work by Zola-Morgan & colleagues



N = Normal Monkeys
 A = Amygdala (A)
 H+ = Hippocampal Formation (HF)
 H+A = HF + A
 H+A+ = large MTL resection
 H++ = HF & Perirhinal Cortex

NMTS-1 = 1 month after surgery
 NMTS-2 = 1-2 years after surgery

Dementias

Loss of cognitive function (including memory) going beyond normal aging

Degenerative Dementias

- Alzheimer's Disease (AD)
- Lewy Body Dementias (DLB & PD)
- Fronto-temporal dementia (FTD)

Vascular Dementias

- multiple strokes
- Etc.

Dementias

Loss of cognitive function (including memory) going beyond normal aging

Degenerative Dementias

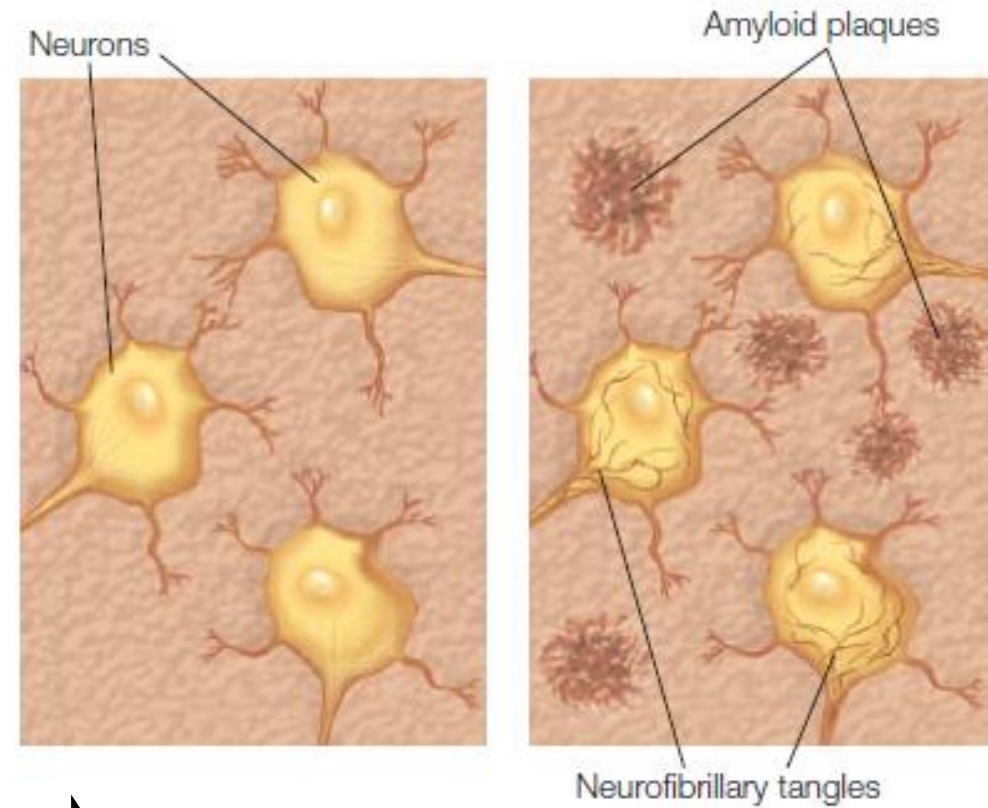
- Alzheimer's Disease (AD)
- Lewy Body Dementias (DLB & PD)
- Fronto-temporal dementia (FTD)

Vascular Dementias

- multiple strokes
- Etc.

Pathophysiology and Anatomy of AD

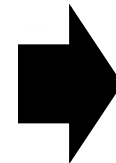
- Neurofibrillary tangles and amyloid plaques.
- Neuronal dysfunction and cell death



Hippocampus and
Entorhinal Cortex



Association cortex



Whole brain

Cognitive Course of AD

Early stage

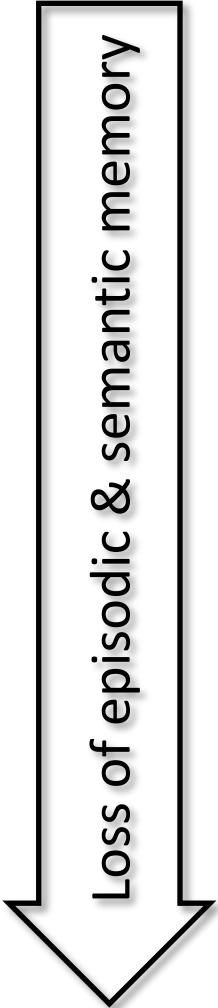
- Forgetfulness (anterograde memory disorders)
- Loss of interest, social withdrawal, difficulties in job, etc.

Middle stage

- Anterograde and retrograde memory disorders (forgetting autobiographic info like birthdays etc.)
- Recognition deficits
- Spatial and temporal disorientation
- „Delusions“

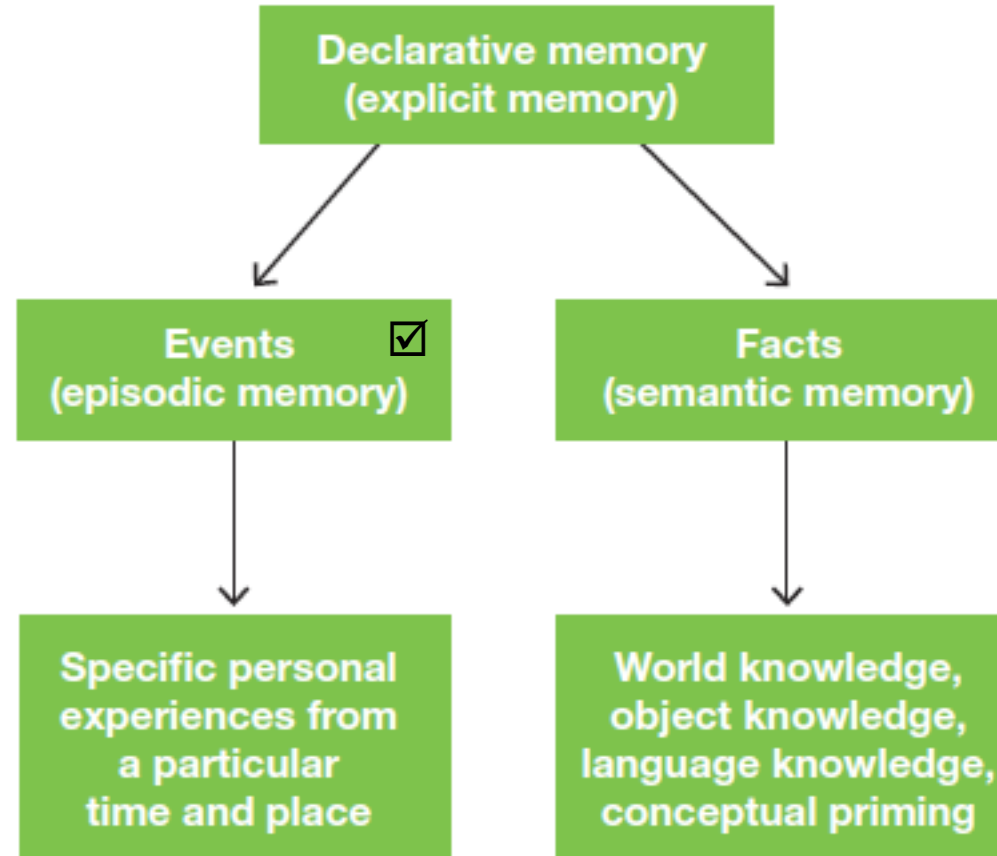
Late stage

- Severe anterograde and retrograde memory disorders
- Severe speech impairments
- Agnosias



Loss of episodic & semantic memory

Events vs. Facts



Dementias

Loss of cognitive function (including memory) going beyond normal aging

Degenerative Dementias

- Alzheimer's Disease (AD)
- Lewy Body Dementias (DLB & PD)
- Fronto-temporal dementia (FTD)

Vascular Dementias

- multiple strokes
- Etc.

Dementias

Loss of cognitive function (including memory) going beyond normal aging

Degenerative Dementias

- Alzheimer's Disease (AD)
- Lewy Body Dementias (DLB & PD)
- Fronto-temporal dementia (FTD)

Vascular Dementias

- multiple strokes
- Etc.

Fronto-Temporal Dementia

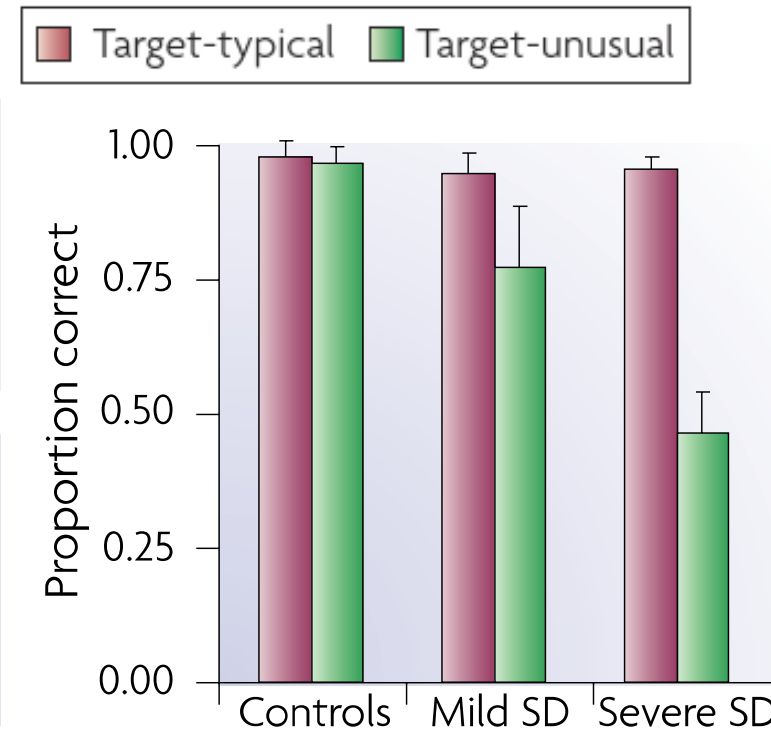
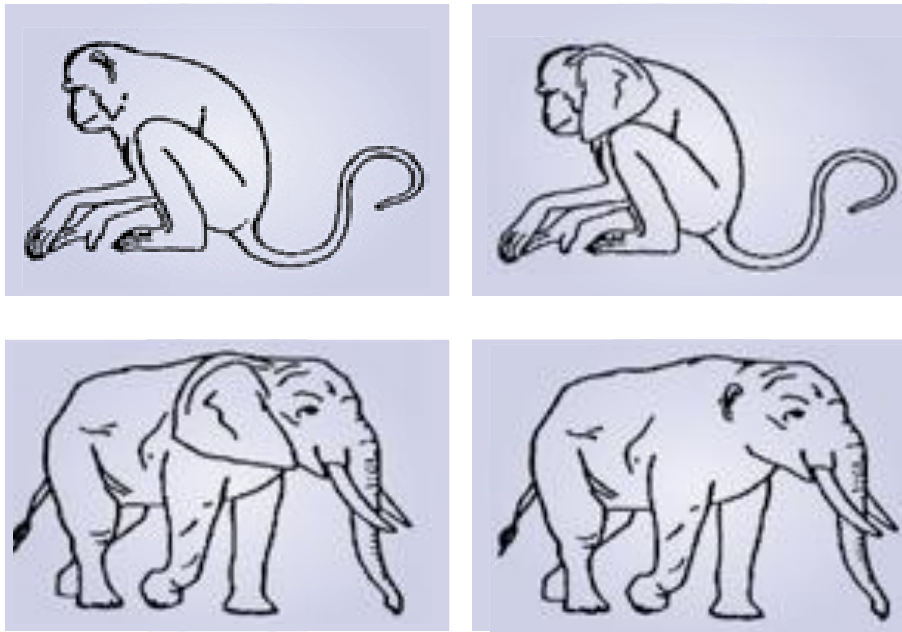
Depending on the localisation and distribution of the degeneration, the symptoms can vary. The heterogenous symptoms are classified in three groups:

- **Behavioural variant FTD:** Disorders mostly in social interactions and control of behaviour.
- **Progressive non-fluent aphasia:** Language disorders are most prominent.
- **Semantic Dementia:** Language is fluent but increasing problems in naming and comprehension.

Semantic Dementia

Although SD is sometimes described as a ,naming' deficit, i.e. a kind of language disorder, patients lose their knowledge of things.

Task two stimulus: Which is the real animal?



Semantic Dementia

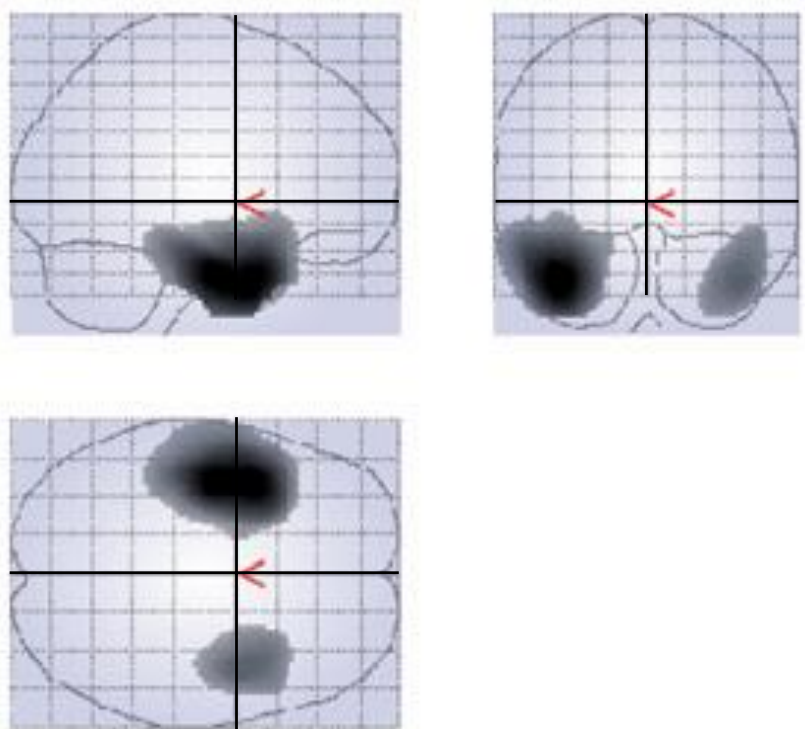
The patients gradually lose their knowledge and concepts of things.
Specific exemplars are replaced by superordinate categories ...

Item	Sept 1991	Mar 1992	Sept 1992	Mar 1993
Bird	+	+	+	Animal
Chicken	+	+	Bird	Animal
Duck	+	Bird	Bird	Dog
Swan	+	Bird	Bird	Animal
Eagle	Duck	Bird	Bird	Horse
Ostrich	Swan	Bird	Cat	Animal
Peacock	Duck	Bird	Cat	Vehicle
Penguin	Duck	Bird	Cat	Part of animal
Rooster	Chicken	Chicken	Bird	Dog

Patterson et al., 2007, Nat Rev Neurosci

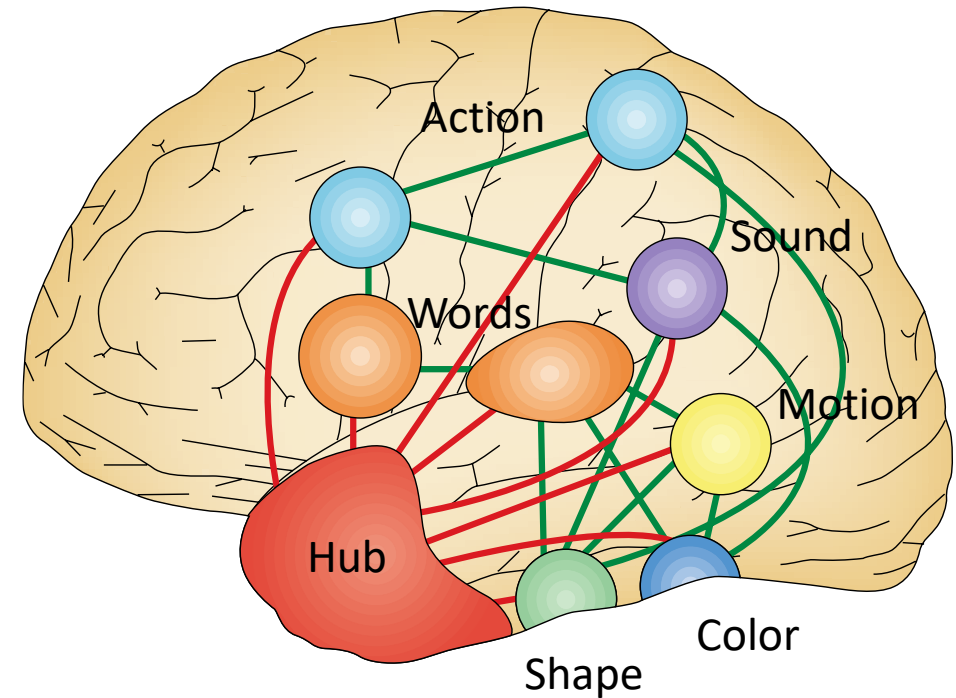
Semantic Memory

Semantic dementia



Areas of reduced metabolism
(also compare Schmolck et al., 2002, Hippocampus)

Distributed-plus-hub view



Patterson et al., 2007, Nat Rev Neurosci

Memory Dysfunction in Dementia

<i>Disease</i>	<i>Episodic memory</i>	<i>Semantic memory</i>	<i>Procedural memory</i>	<i>Working memory</i>
Alzheimer's disease	+++	++	–	++
Frontotemporal dementia	++	++	–	+++
Semantic dementia	+	+++	?	–
Lewy body dementia	++	?	?	++
Vascular dementia	+	+	+	++
Parkinson's disease	+	+	+++	++
Huntington's disease	+	+	+++	+++
Progressive supranuclear palsy	+	+	++	+++
Korsakoff syndrome	+++	–	–	+/-
Multiple sclerosis	+	+/-	?	++
Transient global amnesia	+++	+/-	–	–
Hypoxic-ischemic injury	++	–	–	+/-
Head trauma	+	+	+/-	++
Depression	+	+/-	++	+/-
Anxiety	+	–	–	+/-
Obsessive compulsive disorder	+	–	++	++
Attention deficit hyperactivity disorder			?	+

+++ = Early and severe impairment; ++ = Moderate impairment; + = Mild impairment; occasional impairment or impairment in some studies but not others; – = No significant impairment; ? = Unknown

Summary

Damage to the MTL memory system:

- anterograde episodic amnesia
- time-dependent retrograde episodic amnesia (Ribot's Law)
- spares implicit long-term memory and short-term memory

Damage to the anterior temporal lobe:

- anterograde and retrograde *amodal* semantic amnesia

Additional damage of cortex:

- short-term/working memory impairment
- loss of procedural knowledge
- loss of modality-specific semantic knowledge (cf. agnosias)

Memory and its Disorders



Axel Lindner

a.lindner@uni-tuebingen.de

Section for Neuropsychology, Hertie-Institute for Clinical Brain Research
Neurobiology of Cognition (NoCo Lab), Dept. Psychiatry and Psychotherapy
University of Tübingen